

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

GENEOSCOPY, INC.,
Petitioner,

v.

EXACT SCIENCES CORPORATION,
Patent Owner.

IPR2024-01330
Patent 11,970,746 B2

Before RYAN H. FLAX, DAVID COTTA, and JAMIE T. WISZ,
Administrative Patent Judges.

FLAX, *Administrative Patent Judge.*

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Exact Sciences Corporation (“Patent Owner”) is the owner of U.S. Patent 11,970,746 B2 (Ex. 1001, “the ’746 patent”). Paper 4, 2. Geneoscopy, Inc. (“Petitioner”) filed a Petition requesting an *inter partes* review of claims 1–4 and 12–19 of the ’746 patent. Paper 1 (“Pet.”). Paper 1. Patent Owner’s Preliminary Response identified that claims 1 and 2 of the ’746 patent had been disclaimed. Paper 9 (“Prelim. Resp.”), 2; Ex. 2041 (disclaimer). Therefore, only claims 3, 4, and 12–19 of the ’746 Patent are at issue in this *inter partes* review, which we instituted on February 14, 2024. Paper 10 (“Institution Decision” or “DI”). On May 30, 2025, Patent Owner filed a Response to the Petition (Paper 19, “Response” or “Resp.”), which was followed by Petitioner’s Reply (Paper 22, “Reply”), which was followed by Patent Owner’s Sur-reply (Paper 30, “Sur-reply”). Oral argument was heard at a hearing held on November 20, 2025, and a transcript of the hearing is of record. *See* Paper 38 (“Hr’g Tr.”).

Petitioner bears the burden of proving unpatentability of the challenged claims and this ultimate burden of persuasion never shifts to Patent Owner. *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). As we address the parties’ positions, it is always in the context of this burden. To prevail, Petitioner must prove unpatentability by a preponderance of the evidence. *See* 35 U.S.C. § 316(e) (2018); 37 C.F.R. § 42.1(d) (2023). After considering the parties’ arguments and supporting evidence, as discussed in detail below, we conclude that Petitioner proves by a preponderance of the evidence that the challenged claims of the ’746 patent are unpatentable.

A. REAL PARTIES-IN-INTEREST

Petitioner identifies itself as the real party-in-interest. Pet. 3. Patent Owner identifies itself as the real party-in-interest. Paper 4, 2. Neither party contests or objects to the other's identification in this proceeding.

B. RELATED MATTERS

The parties each identify *Exact Sciences Corp. v. Geneoscopy, Inc.*, No. 24-cv-00583-MN (D. Del.) as involving the '746 Patent, which was consolidated with case No. 23-cv-1319-MN (D. Del.). Pet. 3–4; Paper 4, 2. In IPR2024-00459 (“the -459 IPR”), Petitioner successfully challenged the claims of U.S. Patent 11,634,781 (“the '781 Patent”),¹ which is the parent of the '746 Patent. *See* Ex. 1031 (Petition in that proceeding); Ex. 1087 (Final Decision in that proceeding). Patent Owner has appealed the Final Decision in the -459 IPR to the U.S. Court of Appeals for the Federal Circuit. *See* IPR2024-00459, Paper 40.

C. THE '746 PATENT

The '746 patent, titled “Fecal Sample Processing and Analysis Comprising Detection of Blood,”² was filed as U.S. Application 18/179,945 on March 7, 2023, and claims priority to a series of continuation applications, including U.S. Application 17/936,335 (“the '335 Application”), and U.S. Provisional Application 61/149,581 (“the '581

¹ During the prosecution before the Office leading to the issuance of the '746 patent, Patent Owner filed a Terminal Disclaimer with respect to the '781 patent to overcome a non-statutory, obviousness-type double patenting rejection. Ex. 1019, 8.

² The '746 patent uses both “fecal” and “faecal,” which refer to the same thing (solid body waste), but the former is the American spelling of the term and the latter is the British spelling. We use the American spelling herein.

Provisional”), which was filed on February 3, 2009. Ex. 1001, codes (54), (21), (22), (63), (60), 1:8–19. Thus, the earliest possible effective filing date of the ’746 patent is February 3, 2009. There is no dispute in this proceeding that the ’746 patent is entitled to this priority date and we apply this priority date in our analysis here.

The ’746 patent relates to methods and kits for analysis of fecal samples. *Id.* at 1:31–32. Its Abstract states that the invention relates to:

A method of processing a fecal sample from a human subject comprising combining a first portion of a collected fecal sample with a stabilizing buffer, combining a second portion of the sample with a solution that prevents denaturation or degradation of blood proteins found in a fecal sample. Embodiments comprise testing nucleic acid extracted from the first portion of the fecal sample for an amount of a human nucleic acid, and testing the second portion of the fecal sample for the presence of human blood.

Id. at Abstr. According to the Specification, colorectal cancer (“CRC”) is a leading cause of cancer-related deaths worldwide. *Id.* at 1:41–43. Mass screening of asymptomatic patients is the cornerstone for detecting and eliminating precursor lesions to reduce the risk of CRC. *Id.* at 1:51–55. Colonoscopy is the primary screening test for CRC because of its high sensitivity and specificity and the ability to contemporaneously remove polyps if found. *Id.* at 1:65–2:1.

That procedure, however, is invasive, costly, and has certain risks, such as infection and perforation of the bowel. *Id.* at 2:1–3. Fecal occult blood testing (“FOBT”), which tests for blood in the stool as a CRC screen, is commonly used, and is less invasive and expensive than colonoscopy. *Id.* at 2:4–12. But, because occult blood in stool can be indicative of different gastrointestinal disorders, further testing is necessary to confirm CRC. *Id.*

There are two types of FOBT: guaiac FOBT (“gFOBT”), which detects peroxidase activity of hemoglobin in fecal blood, and immunochemical FOBT (“iFOBT” or “FIT”), which uses anti-human hemoglobin (Hb) antibodies to detect fecal blood. *Id.* at 2:13–34. Although the immunochemical procedure is comparatively more complicated and expensive, iFOBT is more sensitive than gFOBT. *Id.* at 2:25–40.

The Specification explains that developments in testing also look specifically for mutations in DNA that are characteristic of colorectal neoplasia, which are detectable in exfoliated epithelial cells in stool. *Id.* at 2:44–47. The Specification explains that increased DNA methylation is an epigenetic alteration that is common in human cancers. *Id.* at 3:5–7. Aberrantly methylated DNA has also been proposed as a potential tumor marker for CRC detection. *Id.* at 3:7–9.

The ’746 patent explains that, although combined assays for detecting CRC have been described, their approaches target either multiple protein markers or multiple DNA alterations. *Id.* at 3:41–43. According to the Specification, “[t]o date, immunochemical tests and DNA tests for CRC detection have been evaluated and compared on a separate basis only.” *Id.* at 3:43–45.

The ’746 patent states that the invention “aims to improve the positive and negative predictive value and also the sensitivity and specificity of detection of colorectal cancer through non-invasive means.” *Id.* at 6:52–55. Accordingly, the invention is based on a combination of tests for detecting proteins and epigenetic modification markers in the same fecal sample. *Id.* at 6:59–62.

The '746 patent concludes with 19 claims. *Id.* at 45:56–48:36. Petitioner challenges claims 3, 4, and 12–19 of the '746 patent (as noted above, claims 1 and 2 are disclaimed). Claim 3 is illustrative of the challenged claims; it depends from the sole independent claim 1, which was disclaimed, but claim 1's recited subject matter remains incorporated into the dependent claims. Claims 1 and 3 are reproduced below:

1. A method of processing a freshly-collected fecal sample without freezing, the method comprising:

a) collecting a fecal sample from a human subject, wherein the fecal sample is collected at home by the human subject;

b) in a sealable vessel, combining a first portion of the fecal sample with a stabilizing buffer, and sealing the sealable vessel; and

c) in a sealable container, combining the second portion of the fecal sample with a solution that prevents denaturation or degradation of blood proteins found in a fecal sample, and sealing the sealable container.

3. A method of processing a fecal sample, the method comprising:

a) obtaining a pair of portions of a fecal sample collected from a human subject, the pair of portions comprising:

i) a sealed sealable vessel containing a first portion of a fecal sample and a stabilizing buffer; and

ii) a sealed sealable container containing a second portion of a fecal sample and a solution that prevents denaturation or degradation of blood proteins found in a fecal sample,

wherein the pair of portions are obtained by the method of claim 1;

b) extracting nucleic acid from the first portion of the fecal sample;

c) testing nucleic acid extracted from the first portion of the fecal sample for an amount of a human nucleic acid; and

d) testing the second portion of the fecal sample for an amount of a blood protein present in the second portion of the fecal sample.

Ex. 1001, 45:56–46:57, 46:64–47:15. Claims 4 and 12–19 depend, either directly or indirectly, from claim 3. Claims 4, 12, and 13 recite further limitations for testing nucleic acid, and claims 14–19 recite further limitations for testing for an amount of a blood protein. *Id.* at 47:17–18, 48:6–36.

D. PATENT OWNER’S COLOGUARD PRODUCT

For context, according to the evidence of record, the ’746 patent is, as of 2021, one of 59 U.S. patents that Patent Owner indicates cover its Cologuard product, approved for use by the U.S. FDA on August 11, 2014. Ex. 1026, 1; Ex. 1027; Ex. 2026, 4, 10. As of its annual 10-K report in 2014 introducing this Cologuard product, Patent Owner identified that, at that time, it had “intellectual property rights pertaining to sample type, sample preparation, sample preservation, biomarkers, and related methods and formulations,” protected then by 12 issued U.S. patents, as well as (potentially) 30 pending U.S. patent applications. Ex. 2026, 10.

According to Patent Owner’s published “Instruction for Use” for the Cologuard product,

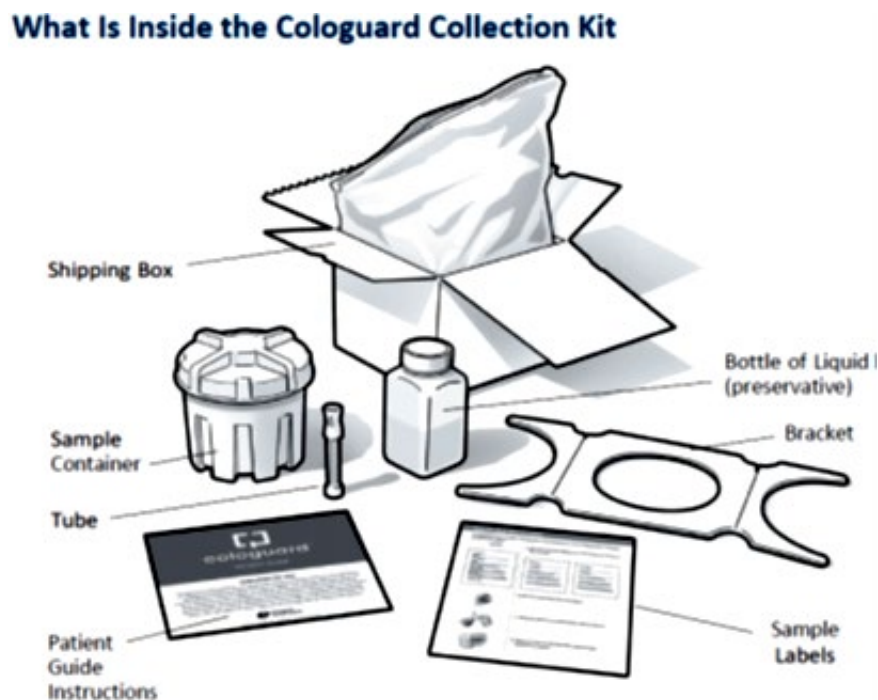
Cologuard is intended for the qualitative detection of colorectal neoplasia associated DNA markers and for the presence of occult hemoglobin in human stool. Cologuard is for use with the Cologuard collection kit and the following instruments: BioTek ELx808 Absorbance Microplate Reader; Applied Biosystems® 7500 Fast Dx Real-Time PCR; Hamilton

Microlab® STARlet; and the Exact Sciences System Software with Cologuard Test Definition.

Ex. 1081, 6. “Cologuard is designed to analyze patients’ stool for the presence of 11 molecular markers, including hemoglobin and DNA markers, which may indicate the presence of colorectal cancer or advanced adenomas. Because cellular exfoliation of DNA into stool occurs continuously, Cologuard can detect pre-malignant neoplasia at early onset of abnormality.”

Ex. 1080, 3; *see also* Ex. 1073; Ex. 1076, 1.

The Cologuard collection kit, referred to above, “includ[es] a stool sample for DNA testing (Container) and a stool sample for Hemoglobin testing (Tube).” Ex. 1081, 15. Patent Owner’s Cologuard Patient Guide provides a labeled illustration of this kit, which we reproduce below:



Ex. 1076, 8. The illustration is titled “What is Inside the Cologuard Collection Kit,” and shows a shipping box, a sample container, a bracket that

appears to be shaped to hold the sample container, a bottle of liquid preservative, a tube, sample labels, and patient instructions. The Patient Guide states that the “bottle of liquid contains a preservative (less than 10% EDTA in Tris buffered solution),” and “[t]he tube contains a 10% albumin in Tris buffered detergent solution with an antimicrobial agent.” *Id.*

E. ASSERTED GROUNDS OF UNPATENTABILITY

Petitioner asserts that claims 3, 4, and 12–19 of the ’746 patent would have been unpatentable on the following grounds:³

Ground	Claim(s) Challenged	35 U.S.C. § ⁴	Reference(s)/Basis
1	3, 4, 14–19	103	Lenhard, ⁵ Vilkin, ⁶ Itzkowitz ⁷
2	12, 13	103	Lenhard, Vilkin, Itzkowitz, Kanaoka ⁸

³ Petitioner’s Ground 5 is omitted here because it is moot in view of Patent Owner’s disclaimer of claims 1 and 2.

⁴ Because the ’746 Patent has an effective filing date before March 16, 2013, the pre-AIA version of § 103 applies. Our decision, however, would be the same under either version of the statute section.

⁵ Lenhard et al., *Analysis of Promoter Methylation in Stool: A Novel Method for the Detection of Colorectal Cancer*, 3 CLIN. GASTROENTEROL. & HEPATOL. 142–49 (2005) (Ex. 1004, “Lenhard”).

⁶ Vilkin et al., *Performance Characteristics and Evaluation of an Automated-Developed and Quantitative, Immunochemical, Fecal Occult Blood Screening Test*, 100 AM. J. GASTROENTEROL. 2519–25 (2005) (Ex. 1005, “Vilkin”).

⁷ Itzkowitz et al., *Improved Fecal DNA Test for Colorectal Cancer Screening*, 5 CLIN. GASTROENTEROL. & HEPATOL. 111–17 (2007) (Ex. 1006, “Itzkowitz”).

⁸ S. Kanaoka, US 2006/0216714 A1, pub. Sept. 28, 2006 (Ex. 1007, “Kanaoka”).

Ground	Claim(s) Challenged	35 U.S.C. § ⁴	Reference(s)/Basis
3	3, 4, 14–19	103	Shuber, ⁹ Vilkin
4	12, 13	103	Shuber, Vilkin, Kanaoka
6	3, 4, 14–19	103	Inbar, ¹⁰ Vilkin, Shuber
7	12, 13	103	Inbar, Vilkin, Shuber, Kanaoka

See Pet. 12. Petitioner relies, *inter alia*, on the Declarations of Duncan Whitney, Ph.D. Ex. 1002; Ex. 1086. Patent Owner relies, *inter alia*, on the Declarations of Vadim Backman, Ph.D. Ex. 2001; Ex. 2044. The qualifications of these witnesses to testify as to the subject matter of their declarations is not disputed in this proceeding.

II. DISCUSSION

A. LEGAL STANDARDS

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)). This burden of persuasion never shifts to Patent Owner.¹¹ See *Dynamic Drinkware*, 800 F.3d at 1378.

⁹ Shuber et al., WO 2005/113769 A1, pub. Dec. 1, 2005 (Ex. 1008, “Shuber”).

¹⁰ Inbar et al., WO 97/25925, pub. July 24, 1995 (Ex. 1009, “Inbar”).

¹¹ Although we may refer to Patent Owner’s arguments as “not persuasive,” this is in the context of assessing the record and the parties’ arguments; we do not shift the ultimate burden from Petitioner.

The Supreme Court in *KSR International Co. v. Teleflex Inc.*, 550 U.S. 398 (2007), reaffirmed the framework for determining obviousness set forth in *Graham v. John Deere Co.*, 383 U.S. 1 (1966). The *KSR* Court summarized the four factual factors set forth in *Graham* (383 U.S. at 17–18) that are applied in determining whether a claim is unpatentable as obvious under 35 U.S.C. § 103 as follows: (1) determine the scope and content of the prior art; (2) ascertain the differences between the prior art and the claims at issue; (3) resolve the level of ordinary skill in the art; and (4) consider objective evidence indicating obviousness or non-obviousness. *KSR*, 550 U.S. at 406.

“The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *Id.* at 416. “[C]ase law is clear that obviousness cannot be avoided simply by a showing of some degree of unpredictability in the art so long as there was a reasonable probability of success.” *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1364 (Fed. Cir. 2007) (citing *In re Corkill*, 771 F.2d 1496, 1500 (Fed. Cir. 1985)). “[W]hen the question is whether a patent claiming the combination of elements of prior art is obvious,” the answer depends on “whether the improvement is more than the predictable use of prior art elements according to their established functions.” *KSR*, 550 U.S. at 417.

KSR explains that “if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill.” *Intel Corp. v. PACT XXP Schweiz AG*, 61 F.4th 1373, 1380 (Fed. Cir. 2023) (quoting *KSR*, 550 U.S. at 417; and *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 797 (Fed. Cir.

2021)). If addressing a known problem, the prior art combination need not be “the *best* option, only . . . a *suitable* option.” *Id.* at 1380 (quoting *Intel*, 21 F.4th at 800 (emphasis original)).”

“[O]bjective indicia of non-obviousness play an important role as a guard against the statutorily proscribed hindsight reasoning in the obviousness analysis,” and “may often be the most probative and cogent evidence in the record.” *WBIP, LLC v. Kohler Co.*, 829 F.3d 1317, 1328 (Fed. Cir. 2016); *see also Graham*, 383 U.S. at 36 (“[Objective indicia] may also serve to guard against slipping into use of hindsight, and to resist the temptation to read into the prior art the teachings of the invention in issue.” (internal quotation marks and citation omitted)). “In order to accord substantial weight to [objective indicia of non-obviousness, also called] secondary considerations[,] in an obviousness analysis, the evidence of secondary considerations must have a nexus to the claims, *i.e.*, there must be a legally and factually sufficient connection between the evidence and the patented invention.” *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019) (citing *Henny Penny Corp. v. Frymaster LLC*, 938 F.3d 1324, 1332 (Fed. Cir. 2019)). “The patentee bears the burden of showing that a nexus exists.” *Id.*

B. ORDINARY SKILL IN THE ART

Petitioner contends that a person of ordinary skill in the art at the time of the invention (which Petitioner calls a “POSA”) would have had “a Ph.D. in chemistry, biochemistry, biology, or a related field and at least five years of experience designing and performing diagnostic assays on fecal samples.” Pet. 12 (citing Ex. 1002 ¶¶ 9–13).

Patent Owner disagrees and asserts that the ordinarily skilled artisan would have had “a doctoral degree in medicine, chemistry, biochemistry, biology, or a related field and 1–2 years of experience in the processing and analysis of biological samples, including fecal samples.” Resp. 20 (citing Ex. 2044 ¶¶ 2–9, 51–53, 56–57). Patent Owner also indicates that such a person could have been an individual or a member of a team of scientists addressing fecal sample processing and analysis. *Id.* Patent Owner notes that Petitioner’s proposal in the Petition is significantly different from the proposal in its *ex parte* reexamination request of the related ’781 patent, which merely required a bachelor’s degree and several years of experience processing biological samples. Resp 20 (citing Ex. 2042, 7 (Request for Reexamination); Ex. 2044 ¶¶ 51–53).¹²

As explained in our Institution Decision (at 7–8) in this proceeding, as well as in our Final Decision in the -459 IPR (*see* Ex. 1087, 12–14), we do not discern much of a substantive difference between the parties’ respective definitions in this proceeding beyond the number of years of experience after obtaining a doctoral degree. *Compare* Pet. 12, *with* Prelim. Resp. 21; *see also* Resp. 20 (“No matter which of the three different levels of skill is applied”). For purposes of this Final Decision, we adopt Patent Owner’s definition as it appears to be reasonable and falls between Petitioner’s definitions asserted in the Petition and in its request for reexamination of the related ’781 patent. Moreover, this definition appears to be consistent with the prior art’s demonstration of the level of ordinary

¹² On occasion, we cite to exhibit page numbers as added by Patent Owner. However, we endeavor herein to cite to the original page numbers of exhibits, where possible.

skill in the art at the time of the invention. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not shown” (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985))).

C. CLAIM CONSTRUCTION

The Board interprets claim terms in an *inter partes* review using the same claim construction standard that is used to construe patent claims in a civil action in federal district court. 37 C.F.R. § 42.100(b). In construing claims here, similar to district courts, we by default give claim terms their ordinary and customary meaning, which is “the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Petitioner does not explicitly address claim construction in the Petition. *See generally* Pet. Similarly, Patent Owner states that it “does not believe that claim construction is necessary to resolve patentability.” Resp. 20. Each party has taken the position in this proceeding that the recited sample collection, sample portioning, and respective combining with stabilizing buffer/solution steps (recited in claim 1) are performed at home. *See, e.g.*, Hr’g Tr. 7:11–8:7, 35:10–19. The patent disclosure supports the parties’ common understanding (*see, e.g.*, Ex. 1001, 10:24–28, 11:51–67, 18:4–19:37, 25:18–47), thus, we adopt the parties’ position.

“[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” *Nidec Motor Corp. v.*

Zhongshan Broad Ocean Motor Co., 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)). Other than our acknowledging and adopting the parties' common position regarding the claimed steps that occur at home, we agree with the parties that no other construction of any claim term is necessary for purposes of our Final Decision.

D. PETITIONER'S ASSERTED PRIOR ART

We review and summarize the relevant disclosures of Petitioner's asserted prior art references below. Petitioner asserts each is prior art and Patent Owner does not contest that any are prior art with respect to the challenged claims. Pet. 11–12; *see generally* Resp.

1. Lenhard (Ex. 1004)

Lenhard is a journal article titled “Analysis of Promoter Methylation in Stool: A Novel Method for the Detection of Colorectal Cancer,” that indicates it was published in the journal *Clinical Gastroenterology and Hepatology* in 2005; it is prior art under 35 U.S.C. § 102. Ex. 1004, 142.

Lenhard identifies that “[c]olorectal cancer (CRC) is one of the leading causes of cancer-related morbidity and mortality,” and “[d]etection of early disease and precancerous adenomatous lesions leads to a decrease of CRC-related mortality. Therefore, there is a strong rationale for screening programs.” *Id.* Of several screening techniques, Lenhard identifies colonoscopy as one, but notes that “[c]ompliance with endoscopic screening is not satisfactory.” *Id.* According to Lenhard, the detection of tumor-derived genetic changes in stool was a promising approach for CRC

screening, and the use of fecal occult blood testing (FOBT),¹³ alone or in combination with endoscopy, was recommended. *Id.* Lenhard states that, “FOBT, when applied regularly, can decrease CRC mortality by only 15–33%.” *Id.*

Lenhard discloses using the gene hypermethylated in cancer 1 (“HIC1”) promoter methylation as a stool-based DNA marker to screen for CRC. *Id.* at 143; *see also id.* (also identifying SFRP2, SFRP5, and PGR as tested markers). According to Lenhard, the promoter of HIC1 frequently is methylated in CRC, but not in normal or aging colonic tissue. *Id.* Lenhard states that it has shown that “HIC1 promoter methylation can be detected frequently and with high specificity in stool samples from patients with CRCs.” *Id.*

Moreover, Lenhard states, “[t]he combination of HIC1 methylation analysis with FOBT allowed for detection of two thirds of CRCs.” *Id.* at 147. According to Lenhard, “[t]he combination of both assays resulted in increased detection rates for CRCs.” *Id.*; *see also id.* at 146 (Table 4) (providing data for positivity rates of HIC1 assay, FOBT assay, and combination of the tests). Lenhard further states that “[a]lthough the

¹³ As discussed above, and as also described in the ’746 patent, there are two types of FOBT identified in the record: gFOBT and iFOBT. Ex. 1005, 2519. gFOBT refers to “guaiac fecal occult blood test.” *Id.* This test is based on a chemical reaction between heme and guaiac, a compound isolated from *Guaiaecum* trees, causing a color change. Ex. 1002 ¶¶ 51–52. iFOBT (sometimes called FIT, for fecal immunochemical test) refers to “immunochemical, fecal occult blood test,” which is “specific for human Hb,” i.e., human hemoglobin; the test is based on antibody detection of hemoglobin. Ex. 1005, 2519; *see also* Ex. 1002 ¶ 53; Hr’g Tr. 22:18–21 (iFOBT and FIT are interchangeable terms).

combined test detected all localized cancers, no increase in sensitivity for adenoma was seen.” *Id.*

Lenhard states that stool samples were collected preoperatively from patients with verified CRCs and before colonoscopy for patients with adenomas larger than one centimeter. *Id.* Samples were received within ten hours after defecation at the laboratory, subjected to gFOBT (color change test) immediately on receipt, and then stored at -80°C until analyzed for methylated DNA. *Id.* at 143, 145.

Lenhard expressly suggests performing FOBT in combination with another screening technique, i.e., colonoscopy and DNA testing. *See generally id.* Upon testing by combining FOBT with gene methylation analysis Lenhard found the combination effective to detect 2/3 of cancer instances. *Id.* at 146–47. The studied FOBT worked as well as reported in the past and the HIC1 was slightly better (more sensitive, detected some other types of cancer better). *Id.* at 147. Using both tests “resulted in increased detection rates.” *Id.* Lenhard praises the combination of FOBT and HIC1 detection as a “highly sensitive, specific, and easily analyzable” process. *Id.* Finally, Lenhard urges the “[e]valuation of other genes known to be methylated in CRC such as hMLH1 and HPP1, [and the] identification of other useful methylation markers.” *Id.* at 148.

2. Vilkin (Ex. 1005)

Vilkin is a journal article titled “Performance Characteristics and Evaluation of an Automated-Developed and Quantitative, Immunochemical, Fecal Occult Blood Screening Test,” that indicates it was published in the *American Journal of Gastroenterology* in 2005; it is prior art under 35 U.S.C. § 102. Ex. 1005.

Focusing on the use of FOBT screening for CRC (as a threshold for conducting colonoscopy), Vilkin explains that gFOBT is faulted for its low sensitivity for significant colorectal neoplasia (i.e., CRC and advanced adenomatous polyps (“AAP”)), and low specificity due to nonspecificity for human hemoglobin (“Hb”). *Id.* at 2519. Vilkin explains that the introduction of central laboratory and office-developed iFOBT, specific for human Hb, improved specificity. *Id.* Vilkin describes a colonoscopy-controlled study for a detailed evaluation of an automated desktop instrument for quantitative, immunochemical determination of fecal occult blood. *Id.* at 2523.

For use in an iFOBT CRC screening procedure, Vilkin describes a fecal test sampling device shaped like a small test tube with a fecal probe that is inserted into the stool and then pushed back into the tube, through a membrane, into a sample cup that includes an Hb stabilizing buffer. *Id.* at 2520. Vilkin’s labeled photograph (Figure 1) of this sampling device and the diagnostic hardware is reproduced below:

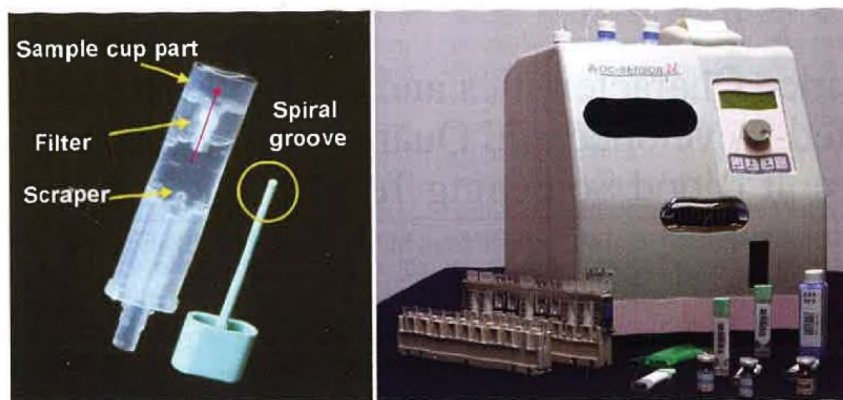


Figure 1. Left: Stool probe and fecal sample storage tube. The closed tube is 8 cm long, 2 cm wide. On each tube there is a unique bar code and space to write name and date. Right: Desktop instrument and accessories. The instrument is 32 cm wide, 53 cm deep, and 42 cm high.

Vilkin’s Figure 1 depicts and labels a fecal sample storage tube (above left) having a sample cup part, a filter, a scraper, and a cap with a prong and a

spiral groove and also shows a desktop instrument (above right) for performing iFOBT analysis (along with several containers on the countertop in front of the instrument). *Id.* Vilkin states:

The fecal test sampling device is shaped like a small test tube with the fecal probe inserted into it and sealing it. The probe has a serrated tip, which is poked into the stool and then pushed back into the tube, past a scraper, and through a membrane into the sample cup. These remove most of the excess feces and leave the stool collected (about 10 mg) into the serrations (Fig. 1). The tip is then put in a closed amount of Hb stabilizing buffer.

Id. Vilkin explains that the samples are double-closed in ziplock bags and kept in the refrigerator until returned to the laboratory where they are kept at 4°C until testing. *Id.* Vilkin explains that “[t]he instrument,” called an OC-Sensor (from Eiken, Japan)¹⁴ and shown in Figure 1, above-right, “is self-contained” and “automatically mixes the fecal buffer solution with the latex-antihuman HbA antibody reagent,” and “is used for developing and quantification of the I-FOBT.” *Id.*

Vilkin discloses that iFOBT screening provides high sensitivity (76.5%) and acceptable specificity (95.3%; better than gFOBT); identifying six of six CRC patients, twenty of 28 advanced adenomatous polyp patients, and 26 of 34 neoplasia patients. *Id.* at 2522–23. Vilkin suggests a need to preserve the sample with buffer, no need to freeze, and stability for shipping to lab (3 weeks just refrigerated). *Id.* at 2523–24.

Vilkin discloses that iFOBT was already proven effective in Japan and Hong Kong, and had the advantages of convenience, cost-reduced screening,

¹⁴ The ’746 patent describes using the same or a similarly named instrument, i.e., “[t]he OC-Hemodia ‘Eiken’ . . . to detect faecal occult blood,” in its Example 1. *See* Ex. 1001, 28:60–62.

and not requiring dietary restrictions. *Id.* at 2519. In Vilkin’s methodology, three iFOBT tests were conducted (3 samples, annually); however, Vilkin discloses that iFOBT collection in Japan, the United States, and Australia is conducted over 2-days (2 samples) annually, and in Italy over 1 day (1 sample) annually. *Id.* at 2524. Vilkin suggests that the use of iFOBT screening, using a sample stability buffer solution and refrigeration (rather than freezing), would be conducted with at-home sample collection in an improved toilet-disposable device and packaging. *Id.*

3. Itzkowitz (Ex. 1006)

Itzkowitz is a journal article titled “Improved Fecal DNA Test for Colorectal Cancer Screening” that indicates it was published in the journal *Clinical Gastroenterology and Hepatology* in 2007; it is prior art under 35 U.S.C. § 102. Ex. 1006. Several of Itzkowitz’s authors were identified as employees of Exact Sciences Corporation. *Id.* at 111.

Itzkowitz explains that several studies have shown the feasibility of detecting colon tumor-specific products (i.e., colorectal cancer or CRC) in patient stool for cancer screening. *Id.* The markers for CRC in these studies represent alterations of genes. *Id.* Itzkowitz teaches that pilot studies showed that several technical and conceptual advances could improve fecal DNA testing sensitivity. *Id.* For example, adding a DNA-stabilizing buffer to the stool immediately after defecation (sample collection) was shown to prevent DNA degradation for several days and enhance the performance of a DNA integrity assay (“DIA”). *Id.* Also, Itzkowitz explains that promoter methylation has become recognized as a key pathway by which colon cancers develop and a marker for CRC. *Id.*

Itzkowitz describes a two-phase study. *Id.* at 112. Phase 1 involved analyzing stool samples from approximately 50 patients with CRC and 200 patients with normal colonoscopy to define suitable DIA cut-off values and to determine optimal markers for the assay. *Id.* Phase 2, which was then ongoing, was designed as a validation set in which an additional 125 patients with CRC and 200 patients with normal colonoscopy would be analyzed using the optimal marker panel from phase 1. *Id.*

Itzkowitz explains that subjects (patients) were given a special stool collection kit to be mounted on a toilet bowl for sample collection. *Id.* Immediately after defecation (in the kit), the subject added 250 ml of a DNA-stabilizing buffer to the stool specimen and then shipped the specimen, at room temperature, overnight, to a laboratory for processing and analyzing. *Id.* Itzkowitz describes this as “a noninvasive screening test that is convenient, safe, and easy to perform at home.” *Id.* at 115. The lab processed and tested “for 22 specific mutations,” performed “DIA integrity assay,” and processed for vimentin and Helicase like Transcription Factor (HLTF).” *Id.* at 112.

Itzkowitz discloses that fecal DNA testing was known to have advantages over existing fecal occult blood tests, that the ability to ship/mail samples for testing removes geographic access barriers, and that if a DNA preservation buffer is used, samples can be preserved, even at room temperature, for shipping to lab. *Id.* at 115. Itzkowitz discloses that the observed improved sensitivity relating to the use of a DNA-preserving buffer and the related techniques was at the expense of a minor decrease in specificity, but that the “improved DNA preservation and extraction” technique was still “a better screening test.” *Id.* at 116.

Itzkowitz discloses that, relating to its DNA sampling and preservation technique, “[i]nherent in the success of any screening test is patient acceptance,” and the authors’ experience was “a very high degree of patient satisfaction.” *Id.* at 115–17 (93% of patients reported they “felt very comfortable” with the testing and 84% indicated they were willing to repeat the test). Itzkowitz concludes that “the assay complexity can be reduced significantly, thereby decreasing cost and facilitating the development of an assay kit that could be managed by local clinical laboratories,” and the goal is “to get patients screened who might otherwise avoid an invasive screening test.” *Id.* at 117.

4. Kanaoka (Ex. 1007)

Kanaoka, titled “METHOD OF DETECTING COLON CANCER MARKER,” is the September 28, 2006, publication of U.S. Application 10/549,389, which was filed on September 19, 2003; Kanaoka is prior art under 35 U.S.C. § 102. Ex. 1007, codes (10), (43), (21), (22), (45).

Kanaoka discloses that its invention relates to the following:

It is intended to provide a non-invasive and convenient method of detecting a tumor marker for diagnosing colon cancer which is superior in sensitivity and specificity to the existing fecal occult blood test. More specifically speaking, a method of detecting a tumor marker for diagnosing colon cancer which comprises collecting biological sample which is immediately frozen using liquid nitrogen in some cases, homogenizing the sample in the presence of an inhibitor of an RNA digesting enzyme to give a suspension, extracting RNA from the obtained suspension, subjecting the extracted RNA to reverse transcription to give cDNA, amplifying the obtained cDNA and then detecting the thus amplified cDNA. This method is

characterized by involving no procedure of separating cell components from the biological sample.

Id. at Abstr. Kanaoka states that, “[t]he tumor marker used in the present invention is COX-2, matrixmetalloprotease (MMP), c-met, CD44 variants, EGF-R, EF-1, Wnt-2, Bradeion, SKP2, KPC-1, KPC-2, PRL-3, Angiogenin, Integrin, Snail, Dysadherin, or the like, and is preferably COX-2.” *Id.* ¶ 44.

Kanaoka discloses that “[t]he biological samples of the present invention . . . preferably are human feces,” and “[t]he biological samples of the present invention can be used as they are, or, in some cases, after frozen.” *Id.* ¶¶ 40–41. Kanaoka recognizes that RNA is less stable than DNA, and so discloses the use of RNase inhibitors to stabilize RNA and the use of liquid nitrogen for sample freezing for preservation. *Id.* ¶¶ 11, 39–43, claims 1, 2, 7, 8.

In one example, Kanaoka discloses that feces samples were obtained, separated into 1 g parts in 5 ml tubes, and then frozen. *Id.* ¶ 57. These samples were tested to measure human hemoglobin (Hb) by immunological fecal occult blood testing to obtain a comparison reference value. *Id.* Next, after homogenization, RNA was extracted from the samples and cDNA transcribed, and then amplified by PCR. *Id.* ¶¶ 57–58. Then, using primers designed for COX-2, CEA (carcinoembryonic antigen protein) and COX-2 were detected. Each of CEA, COX-2, and blood via the iFOBT were detected in patients with colon cancer (almost every time). *Id.* ¶¶ 63–66.

5. Shuber (Ex. 1009)

Shuber, titled “Method for Stabilizing Biological Samples for Nucleic Acid Analysis,” is the December 1, 2005, published version of International Application PCT/US2005/017046; Shuber is prior art under 35 U.S.C.

§ 102. Ex. 1009, codes (12), (21), (43), (54). Shuber indicates its applicant is Exact Sciences Corporation. *Id.* at code (71).

Shuber states, “[d]isclosed is a method for preparing nucleic acid containing biological samples for a nucleic acid integrity assay and/or multiple mutation analysis by incubating the biological sample with a stabilization solution that includes a buffer, a chelating agent and a salt.” *Id.* at Abstr. According to Shuber, such a method is used “for colon cancer detection from stool.” *Id.* at 6:9–11.

Shuber discloses methods for preparing nucleic acid-containing biological samples for an assay to detect nucleic acid markers indicative of cancer. *Id.* at 1:11–13. According to Shuber, contacting a patient sample with a stabilization solution stabilizes the DNA so that intact nucleic acids indicative of diseased cells are more effectively detected in a nucleic acid integrity assay. *Id.* at 2:7–9. Shuber explains that a stabilization solution “may be particularly useful when samples are not refrigerated or frozen” and if a sample “is obtained at a remote location and mailed or delivered to a testing center.” *Id.* at 10:10–14; *see also id.* at 7:10–21.

In one aspect of the Shuber invention, a stool sample may be directly deposited into a sealable container and a stabilization solution may be added to the container, after which the container may be sealed for storage/shipping to a test lab. *Id.* at 29:8–19. Shuber also discloses “a sample (e.g., a stool sample) may be directly deposited into a container (e.g., a sealable con[tainer]) that already contains stabilization solution. The amount of solution may be sufficient to cover the sample or a portion thereof.” *Id.* at 30:8–11.

6. Inbar (Ex. 1009)

Inbar, titled “Collection and Transportation of a Pasty Material Sample,” is the July 24, 1997, published version of International Application PCT/IL97/00017, filed on January 13, 1997; it is prior art under 35 U.S.C. § 102. Ex. 1009, codes (43), (54).

Inbar relates to a system for collecting, storing, and transporting “pasty” material, such as stool. *Id.* at Abstr. Inbar discloses a system with multiple separate sealable containers, each containing unique preservation media corresponding to laboratory assays. *Id.* at Abstr., 1:2–12, 3:48, 4:18–5:22; 9:7–30, Figs. 1–4; *see also* Ex. 1002 ¶¶ 138–142). Inbar allows for the specimen to be preserved and sealed within each container, for an extended time, to allow for shipment to a laboratory for testing. *See* Ex. 1009, Abstr., 3:18–32.

To achieve different assays of the same sample material, which likely requires different media, Inbar describes at least two containers each holding a different preservation medium corresponding to each assay. *Id.* at 4:13–18. Further, Inbar explains the containers are sealed by both a pierceable membrane and a closure arrangement for sealing the container after sample collecting member insertion. *Id.* at 8:1–25; Fig. 1.

Inbar shows a two-container apparatus at Figure 1, reproduced below:



Fig. 1

According to Inbar, “Fig. 1 is an exploded view of a system of the invention comprising two containers and a sampling device with two corresponding collecting members.” *Id.* at 7:18–20. Figure 1 shows system 10 having two containers 12, 12', base member 14, receptacles 16, 16', sampling device 20 with handle 22 and two collecting members 24, 24'. *Id.* at 8:2–8.

Containers 12, 12' hold media 26, 26', which may be the same or different, for preserving the samples (e.g., bacteria or parasites). *Id.* at 8:9–11.

Collecting members 24, 24' are used for inserting into a sample, e.g., stool, where the sample is retained thereon; thereafter the collecting members 24, 24' are inserted through membranes 28, 28' into containers 12, 12', which are closed for shipping to a laboratory. *Id.* at 9:7–30.

Although Inbar focuses most on the preservation of microorganisms, it also indicates that stool may be collected for “detection of cryptic blood, [and the] preparation of a biochemical profile.” *Id.* at 1:7–10. The stool may itself be collected using “a collecting member for collecting an excreted stool sample” with “attachment of members for attachment to a toilet seat.” *Id.* at claim 12, Fig. 7.

E. OBJECTIVE INDICIA OF NON-OBVIOUSNESS

This decision ultimately considers all *Graham* factors, but, because each of Petitioner’s Grounds for unpatentability invokes the same objective indicia considerations argued by Patent Owner, we address those considerations at the outset. In opposition to each of Petitioner’s Grounds for unpatentability, Patent Owner argues that objective indicia of non-obviousness (which Patent Owner refers to as “secondary considerations”), in the form of commercial success, failures of others, long-felt and unmet need, industry praise, and copying, provide evidence of the claims’ patentability. Resp. 51–64.

Petitioner argues, *inter alia*, that Patent Owner’s position on objective indicia of non-obviousness fails because Patent Owner does not establish that a nexus exists between the evidence and the challenged claims. Reply 21–26.¹⁵

¹⁵ Here, as explained below, and as in related IPR2024-00459, Petitioner is correct. See Ex. 1087, 23–34 (Final Decision in IPR2024-00459 addressing the issue). In the related -459 IPR, oral argument was heard on this issue on May 1, 2025. See IPR2024-00459, Paper 38 at 21:23–31:26, 49:9–53:3, 67:2–9. Patent Owner’s Response in this proceeding was filed almost a month after that hearing, which addressed the issues of objective indicia of non-obviousness, nexus, and the evidence of record (*id.*); Patent Owner’s Response here presents essentially the same arguments and evidence as in

1. Nexus

As noted above, a nexus is critical to objective indicia of non-obviousness, and the Federal Circuit instructs that “[t]he patentee bears the burden of showing that a nexus exists.” *Fox Factory*, 944 F.3d at 1373 (quoting *WMS Gaming Inc. v. Int’l Game Tech.*, 184 F.3d 1339, 1359 (Fed. Cir. 1999)). And, the Federal Circuit further instructs that, “[t]o determine whether the patentee has met that burden, we consider the correspondence between the objective evidence and the claim scope.” *Id.* (quoting *Henny Penny*, 938 F.3d at 1332).

A rebuttable presumption of nexus between the asserted evidence of objective indicia of non-obviousness and a patent claim is established if the patent owner shows that evidence is tied to a specific product, which “*is the invention disclosed and claimed*,” i.e., the product “embodies the claimed features, and is coextensive with them.” *Demaco Corp. v. F. Von Langsdorff Licensing Ltd.*, 851 F.2d 1387, 1392 (Fed. Cir. 1988) (emphasis added); *Polaris Indus., Inc. v. Arctic Cat, Inc.*, 882 F.3d 1056, 1072 (Fed. Cir. 2018) (quoting *Brown & Williamson Tobacco Corp. v. Philip Morris Inc.*, 229 F.3d 1120, 1130 (Fed. Cir. 2000)). “Although we do not require the patentee to prove perfect correspondence to meet the coextensiveness

the prior proceeding, which we found to be unpersuasive. *Compare* Resp. 51–64 (and Patent Owner’s cited evidence: Ex. 1002 ¶¶ 344–345; Ex. 1006; Ex. 1020; Ex. 1029; Ex. 1042; Ex. 1076; Ex. 2019 ¶¶ 21–26, 29–32, 36–39; Ex. 2021; Exs. 2024–2025; Ex. 2029; Ex. 2031; Exs. 2033–2040; Ex. 2044 ¶¶ 29–39, 196, 199–213, 216–221, 225–228, 230–231, 233–235, 238–246; Ex. 2051) with IPR2024-00459 Paper 20 at 50–66 (PO Response and Patent Owner’s cited evidence: Ex. 1006; Ex. 1024; Ex. 1043; Ex. 2022 ¶¶ 29–36, 39, 143–157, 167, 169–170, 172–174, 177–185; Ex. 2023 ¶¶ 21–26, 29–32, 35–39; Ex. 2024 ¶¶ 52–66; Ex. 2029; Exs. 2031–2032; Exs. 2034–2040; Exs. 2042–2046; Exs. 2048–2050; Exs. 2052–2053).

requirement, what we do require is that the patentee demonstrate that *the product is essentially the claimed invention.*” *Fox Factory*, 944 F.3d at 1374 (emphasis added).

Patent Owner points to the Cologuard product for the nexus required of its objective indicia of non-obviousness evidence. Resp. 52–56; *see also supra* Section I.D (discussing this product). More specifically, Patent Owner (as in IPR2024-00459) walks through the disclosure of the Cologuard Patient Guide (Ex. 1076) and maps it to the limitations of claims 1 and 3 to establish that the product embodies the claims and is coextensive therewith. Resp. 52–56 (citing, *inter alia*, Ex. 1076, 8, 24–28, 30–31; Ex. 2044 ¶¶ 199–213). Patent Owner argues that the claimed invention “is not a subcomponent of the entire product—it is Cologuard’s fundamental workflow.” *Id.* at 52.

Although Patent Owner does not overlook that “Cologuard [is] covered by other patents,” Patent Owner’s position is simply not supported in view of the record’s evidence of dozens of other U.S. patents that Patent Owner lists as covering the Cologuard product and Patent Owner’s express statements to the Office that some of these patents’ (or, then, applications’) claims covering subject matter patentably different than that of the claims here, are critical to Cologuard. *Id.* at 54–55; *see also* Exs. 1026, 1027 (Patent Owner’s listing of Cologuard patents); Ex. 1029, 7–9 (Patent Owner advocating that other patent/application claims covering subject matter critical to FDA approval and Cologuard’s success where others failed); *see also infra* at 32–33 (reproducing the list of Cologuard patents). We discuss this further below.

As one example, regarding commercial success, Patent Owner asserts that Cologuard sales have grown over the years to the millions of tests and hundreds of millions and, now, billions of dollars in revenue. Resp. 56–58 (citing, e.g., Ex. 2019 ¶¶ 29–31, 36–38; Ex. 2021, 11; Ex. 2044 ¶¶ 216–221). Regarding the failures of others and the satisfaction of a long-felt and previously unmet need, Patent Owner points to CRC screening testing available (or not) before Cologuard, the introduction of Cologuard, and its success. *Id.* at 58–61 (citing, *inter alia*, Ex. 2044 ¶ 231). As for evidence of industry praise, again Patent Owner identifies that Cologuard has repeatedly been the subject of awards and praise. *Id.* at 62–63 (citing, *inter alia*, Ex. 2019 ¶ 26). Finally, regarding copying, Patent Owner asserts that Petitioner’s “ColoSense Collection Kit contains the same elements as Exact Science’ successful Cologuard Collection Kit.” *Id.* at 63–64 (citing, *inter alia*, Ex. 2044 ¶¶ 238–246). Thus, all of Patent Owner’s evidence of objective indicia of non-obviousness and arguments thereover relies on the Cologuard product embodying the challenged claims.

We have no reasonable doubt that *aspects* of the Cologuard product practice the invention of the ’746 patent’s claims 1 and 3, which are most generally directed to steps of collecting two portions of a fecal sample at home, and combining a respective buffer stabilizer/preservative with each of the portions for claim 1, and then processing those fecal samples by testing for blood and nucleic acids (e.g., DNA) for claim 3 (without freezing). These appear to be nearly the most basic steps for using the Cologuard kit. *See generally* Ex. 1076. And, these steps do illustrate a certain ease of use and consumer-friendliness that may be a contributing factor in Cologuard’s success. *See, e.g.*, Ex. 2044 ¶¶ 200–202, 214–215, 219.

However, Petitioner argues that a nexus cannot be presumed to exist between the claims and the Cologuard product and, furthermore, that Patent Owner has not otherwise demonstrated a nexus, because Cologuard is “clearly not coextensive with the claimed method.” Reply 21–22 (citing Ex. 1086 ¶¶ 129–132). Petitioner argues that Cologuard is a multi-faceted product of which the components and processes covered by the ’746 patent are but a part, which is decisively illustrated by the fact that Patent Owner has marked the Cologuard product with dozens of other patents. *Id.* at 22–23 (citing Ex. 1002 ¶¶ 335–347; Ex. 1086 ¶¶ 130–131; Ex. 1077, 1295; Ex. 1079); *see infra* at 33–34 (listing such patents).

Petitioner identifies that Patent Owner has publicly attributed Cologuard’s diagnostic success “to the DNA marker and algorithm components of the test,” which is not part of the ’746 patent’s claims. *Id.* at 24 (citing Ex. 1077, 1295). Petitioner also identifies that Patent Owner itself has expressly stated that Cologuard’s success can be attributed “to a later-invented ‘combination of markers and methods for extracting the marker DNA,’” which is also not claimed in the ’746 patent. *Id.* (citing Ex. 1029, 9 (Patent Owner arguing to the Office in an “After Final Amendment” dated July 31, 2015 in U.S. application 14/145,082, which issued as US 9,163,278 B2 (“the ’278 patent”)¹⁶ on Oct. 20, 2015, that these separately-claimed inventions were the reasons for Cologuard’s success and satisfaction of the long-felt need)).

We agree with Petitioner and are persuaded by its arguments.

¹⁶ Office records show the ’278 patent includes claims to “[a] system for isolating a target human DNA from a human stool sample.”

The Federal Circuit addressed facts similar to those before us here in *Fox Factory*, which dealt extensively with the issue of nexus and objective indicia of non-obviousness (which the Federal Circuit there called “secondary considerations”). *Fox Factory*, 944 F.3d at 1372. At the outset, the Court identified that “[i]n order to accord substantial weight to secondary considerations in an obviousness analysis, ‘the evidence of secondary considerations must have a ‘nexus’ to the claims, i.e., there must be ‘a legally and factually sufficient connection’ between the evidence and the patented invention.’” *Id.* at 1373 (quoting *Henny Penny*, 938 F.3d at 1332). The *Fox Factory* decision also addressed coextensiveness, a question of fact, stating that it is the cornerstone of the nexus requirement, that it is rarely shown by “perfect correspondence between the claimed invention and the product,” but that nexus is not present where “the patented invention is only a component of a commercially successful machine or process.” *Id.* at 1373–74.

Fox Factory involved *two* patents, each with claims directed to aspects of bicycle chainrings (i.e., the gears turned by a bike’s pedals) and the configuration of teeth thereon, and a product called X-sync asserted to be the embodiment of the invention(s). *Id.* at 1369–70. The record of *Fox Factory* evidenced that the X-sync product indisputably included features that were “critical” and “more than an insignificant feature,” but not claimed in the first patent. *Id.* at 1374–75. These features of X-sync, unclaimed in that first patent, were touted by patentee as claimed in the second patent. *Id.* The patentee, in two separate proceedings before the Board over these two separate patents, had held up the X-sync product as evidencing objective

indicia of non-obviousness for separate features of the product, which were claimed separately in the two different patents.

The Federal Circuit held that there was no nexus on such evidence (although the patentee would be given a chance to prove nexus on remand). *Id.* at 1378. In so deciding, the Federal Circuit held that when there are features not claimed in a patent that materially impact the functionality of the product, nexus is not presumed. *Id.* at 1376. In later decisions, the Federal Circuit has clarified that *critical* unclaimed features are of particular importance. *See Teva Pharms. Int’l GmbH v. Eli Lilly and Co.*, 8 F.4th 1349, 1362–63 (Fed. Cir. 2021). Moreover, when analyzing the coextensiveness required for nexus, it cannot be reduced to “an inquiry into whether the patent claims broadly cover the product;” the product must *be* “the invention disclosed and claimed,” and *not* “only a component of a commercially successful machine or process.” *Fox Factory*, 944 F.3d at 1377 (quoting *Demaco*, 851 F.2d at 1392).

Fox Factory involved evidence of only *two* patents covering a single product. Here, there are *dozens* of patents, in addition to the ’746 patent, identified as covering Cologuard. *See* Ex. 1026 (Exact Sciences’s website marking Cologuard patents). Exhibit 1026 states:

The Cologuard® test and its use are covered by one or more of the following patents, associated pending patent applications, or non-U.S. international equivalents: U.S. 8,361,720, 8,673,555, 8,715,937, 8,722,330, 8,916,344, 8,969,046, 8,980,107, 8,993,341, 8,999,176, 9,000,146, 9,024,006, 9,057,098, 9,121,070, 9,121,071, 9,127,318, **9,163,278**, 9,169,511, 9,211,112, 9,212,392, 9,290,797, 9,315,853, 9,376,721, 9,399,800, 9,631,228, 9,637,792, 9,657,330, 9,750,482, 9,783,856, 9,803,249, 9,845,491, 9,856,515, 9,982,308, 10,000,817, 10,047,390, 10,093,987, 10,106,844, 10,138,524, 10,144,953, 10,196,676, 10,253,358, 10,265,054, D849,266,

10,435,753, 10,450,614, 10,486,158, 10,519,510 10,590,489, 10,604,793, 10,626,465, 10,702,250, 10,704,083, 10,808,286, 10,822,639, 10,870,893, 10,889,867, 10,894,256, 11,634,781, 11,845,991, and 11,970,746. This product and its use are also covered by patents pending in the U.S. and around the world.

Ex. 1026, 1 (emphasis added to highlight the '746 patent challenged here, the '781 patent challenged in IPR2024-00459, and the '278 patent identified by Patent Owner as claiming particularly important elements of the Cologuard Product).

Furthermore, during prosecution of the '278 patent (see list above), Patent Owner expressed that elements of Cologuard not covered by the '746 patent or the '781 patent, but covered by the '278 patent (i.e., “a particular combination of markers and methods for extracting the marker DNA,” “particular combination of ‘genes of interest,’” the “particular combination of reagents,” and also, potentially (there is no direct evidence this is covered by another patent), the diagnostic accuracy provided by “the DNA marker and algorithm components of the test”), are reasons for Cologuard’s successes where others failed.¹⁷ See Ex. 1029, 6–9 (note, this exhibit at page 7 cites the following Exhibit 2008 as evidence supporting Patent Owner’s prosecution arguments); Ex. 1077, 1294–95; *see also* IPR2024-00495, Paper 38 (Hr’g Tr.) 44:12–45:4 (Patent Owner argued at that Final Hearing that

¹⁷ Patent Owner expressed that these unclaimed (by the '746 patent) features “addressed [the] long-felt need and succeeded where others had failed, as evidenced by the first-ever FDA approval for the product containing these components.” Ex. 1029, 9. The fact that other patented inventions (of the '278 patent), not at issue in this proceeding, met the long-felt need and were the reason for success (of Cologuard) supports our giving little weight to Patent Owner’s arguments on long-felt need and failures of others/success of Patent Owner.

“detection of adenomas . . . is a critical feature” of Cologuard—one not recited in any claim of the ’746 patent).

Thus, even though *aspects* of the Cologuard product appear to embody claims of the ’746 patent, the product has features not covered by the ’746 patent’s claims, some of which are critical unclaimed elements according to Patent Owner. Accordingly, Cologuard is not coextensive with the challenged claims, and no nexus between the product and the challenged claims is shown on this record.

Other than mapping claims 1 and 3 to aspects of Cologuard, which we explained above is not indicative of the complete product and all features identified by Patent Owner as important or critical to the product, and asserting that Cologuard embodies such claimed features to establish a nexus, Patent Owner does not substantively address its many other patents that purportedly cover Cologuard, or its prior statements to the Office regarding important elements of Cologuard covered by the claims of other patents. Resp. 51–64; Sur-reply 20–26.

Although there is no nexus established here, for completeness we address Patent Owner’s specific asserted objective indicia arguments below. Patent Owner asserts commercial success, satisfaction of a long-felt need (others failed), praise, and copying (Resp. 55–66), and we discuss below the flaws in Patent Owner’s positions, including why Patent Owner has not persuasively established coextensiveness with the claims of the ’746 patent sufficient to establish a nexus for these indicia.

2. Commercial Success, Long-Felt Need (and Failures), Praise, and Copying

Where a party is not entitled to a presumption of nexus, it bears the burden of showing that the success of the product is “a direct result of

unique characteristics of the claimed invention.” *Fox Factory, Inc.*, 944 F.3d at 1373–74. Patent Owner does not show here that success is the “direct result of unique characteristics of the claimed invention.” For each of the alleged objective indicia, we address whether the evidence supports the existence of a nexus even absent a presumption.

Regarding commercial success, Patent Owner asserts that the billions in revenue from sales of Cologuard (through 2023), due to the “diagnostic accuracy and uptake” of the product, are reflected in the ’746 patent’s claims, particularly the recited addition of buffers to the sample portions and the recited “at-home collection,” and are evidence of the non-obviousness of the challenged claims. Resp. 56–58. We agree that Cologuard sales have been substantial. *See, e.g.*, Ex. 2021, 11. However, the record does not support that Cologuard’s sales are *due to* the invention claimed in the ’746 patent.

Petitioner points out that the Cologuard product had and has market exclusivity (provided by, e.g., other patents and FDA approval), and that Patent Owner has publicly stated that it spent over \$3 billion on sales and marketing for Cologuard. Reply 25–26 (citing Ex. 2019 ¶ 13); *see also* Ex. 1029, 9 (identifying Cologuard as the “first-ever FDA approval for the product containing . . . components” claimed by another patent). Patent Owner itself argues that it embarked on an education campaign for Cologuard, where it “expended substantial resources on this education effort with marketing focused on the availability, advantages, and ease of use of Cologuard”; Patent Owner, however, does not account for how influential the substantial investment in education and marketing was in the success of Cologuard compared to its features claimed in the ’746 patent. *See* Resp. 58

(citing Ex. 2019 ¶¶ 31–32); *see also* Hr’g Tr. 48:1–4 (“JUDGE FLAX: I mean, to be fair, your client has sold billions of dollars worth of a product that requires a scraping, right? MS. HABERNY: Well, part of those sales are, you know, a result of education and things like that.”). “[T]here may be many reasons a product is commercially successful; it is only where the success is due to the claimed invention that commercial success can show nonobviousness,” and here there is evidence that factors other than the ’746 patent’s claimed invention may have contributed (and likely did contribute) to Cologuard’s commercial success. *UCB, Inc. v. Actavis Labs. UT, Inc.*, 65 F.4th 679, 695 (Fed. Cir. 2023). Patent Owner has not established that Cologuard’s commercial success is due to the ’746 patent’s claimed invention, rather than other factors as described above (e.g., a particular combination of markers and methods for extracting the marker DNA, or education and marketing) and, for this reason as well, we give the evidence of commercial success little weight.

Regarding the asserted failures of others and long-felt need, Patent Owner argues that the accessibility of the Cologuard product for patients satisfied a need for “a CRC screening test with sufficient diagnostic accuracy that would be used by patients,” and that “low screening levels of individuals within the recommended screening groups persisted until the introduction of Cologuard.” Resp. 58–60 (citing, *inter alia*, Ex. 2044 ¶¶ 29–39, 226–228, 231). Patent Owner argues that Cologuard met these needs “due to [its] non-invasive nature and lack of complications,” which addressed “all major limitations of the other attempts at CRC screening.” *Id.* at 60 (citing, *inter alia*, Ex. 2019 ¶¶ 37, 39; Ex. 2044 ¶ 231). The Cologuard product appears to meet the above needs, and appears to have been accepted

and purchased by a large population of patients, as asserted by Patent Owner, but there is no persuasive evidence that this is due to *the* invention claimed in the '746 patent rather than the inventions covered by the dozens of other Cologuard-protecting patents, including the ***diagnostic accuracy*** attributed to the DNA marker and algorithm components, and methods for extracting DNA, claimed ***in the '278 patent***, which appear to at least contribute to meeting the alleged long-felt need. *See* Ex. 1029, 6–9; Ex. 1028 (the '278 patent); Reply 24–25; *see also* Ex. 2008, 1295 (diagnostic accuracy attributable to DNA markers and algorithm of test). For these reasons, we give this evidence little weight.

As for the asserted industry praise, Patent Owner identifies several instances where Cologuard has won awards or earned public praise, for example, as the best medical technology product of 2016, the “best of what’s new” in 2014, and as an “emerging tech” in 2015. Resp. 62–63. But, other than recognizing the Cologuard product *generally* as innovative and praiseworthy, it is not apparent from the record that such praise can be attributed to *the invention* of the challenged claims rather than some other element(s) of Cologuard covered by its dozens of other patents, as discussed above. *See* Exs. 2033–2035. For this reason, also, we give this evidence little weight.

Finally, Patent Owner has identified that Petitioner has developed a product called ColoSense that Patent Owner alleges copies “the same elements as Exact Sciences’ successful Cologuard Collection Kit.” Resp. 63. At a high level, the ColoSense kit appears similar to Cologuard because it appears to be intended for providing a fecal sample at home for CRC screening (by detecting nucleic acids and blood) and the ColoSense kit

has containers for two portioned parts of the sample with respective preservatives/buffers.¹⁸ *Id.* at 64–65 (citing Ex. 2053, 37–40). On this record, it is unclear whether the asserted copying is directed to the challenged claims, the claims of other Cologuard patents, or some combination thereof. Because Patent Owner’s Cologuard cannot be said to be coextensive with the challenged claims, evidence of copying Cologuard, even if taken as true, cannot be given substantial weight because of the same nexus issues discussed above. Moreover, “copying ‘requires evidence of efforts to replicate a specific product,’” and even *established* “infringement, taken alone, is not probative of copying.” *Tokai Corp. v. Easton Ent., Inc.*, 632 F.3d 1358, 1370 (Fed. Cir. 2011) (quoting *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1246 (Fed. Cir. 2010); *Iron Grip Barbell Co., Inc. v. USA Sports, Inc.*, 392 F.3d 1317, 1325 (Fed. Cir. 2004)). Similarity between products, which is the only evidence of record with respect to copying, does not establish copying. For these reasons, we give this evidence little weight.

3. Summary and Conclusion

For the reasons discussed above, and because the nexus issue affects each basis for objective indicia of non-obvious asserted by Patent Owner, we give no substantial weight to any of Patent Owner’s arguments on such indicia. *In re Affinity Labs of Texas, LLC*, 856 F.3d 883, 901 (Fed. Cir.

¹⁸ This comparison between Cologuard and ColoSense relies on what, at this point, can be called an original ColoSense product design, as set forth in Exhibit 2040. The record indicates that this original design has been changed in favor of a new product design having “a single-container, single buffer,” rather than separate containers for two sample portions with respective preservative buffers. Sur-reply 4 (citing Ex. 2057, 30:4–31:18); *see also* Hr’g Tr. 47:13–25 (Patent Owner discussing a new version of ColoSense having a single container, not requiring separation of the sample).

2017) (without an established nexus between the claimed invention and asserted objective indicia of non-obviousness, such arguments are accorded little weight).

In any event, even if any of the alleged objective indicia of non-obvious were entitled to some weight, after analysis of all the obviousness considerations on the complete record as discussed below, we find that the evidence of obviousness on this record is quite strong and substantially outweighs any indicia of non-obviousness on this unique record. *Pfizer*, 480 F.3d at 1372 (“Although secondary considerations must be taken into account, they do not necessarily control the obviousness conclusion.”).

F. OBVIOUSNESS OVER LENHARD, ITZKOWITZ, AND VILKIN, AND ALSO LENHARD, ITZKOWITZ, VILKIN, AND KANAOKA (GROUNDS 1 AND 2)

Grounds 1 and 2 are very similar, each being premised on substantially the same prior art combination and party arguments for and against patentability. Therefore, we address them together. Petitioner contends, under Ground 1, that claims 3, 4, and 14–19, would have been unpatentable as obvious over the combination of Lenhard, Vilkin, and Itzkowitz. Pet. 19–34 (citing Ex. 1001, 12:48–52; Ex. 1002 ¶¶ 144–219; Ex. 1004, 142–45, 147; Ex. 1005, 2519–24; Ex. 1006, 111–12, 115–17; Ex. 1020, 29–30; Ex. 1021, 214; Ex. 1022, 186, 190; Ex. 1023, 140; Ex. 1033, 9–20 (IPR2024–00459, Paper 9 (Institution Decision))); *see also supra* Section II.D (discussing the asserted prior art). And, under Ground 2, Petitioner argues that claims 12 and 13 would have been unpatentable over the combination of Lenhard, Itzkowitz, Vilkin, and Kanaoka. Pet. 35–37 (citing, *inter alia*, Ex. 1002 ¶¶ 220–226; Ex. 1007 ¶¶ 21–26, 47–48, 55–68, 76; Ex. 1024, 131).

Patent Owner opposes these challenges. Resp. 1–2, 9–16, 21–36, 47–64 (citing, *inter alia*, Ex. 1004; Ex. 1005, 2519–24; Ex. 1006, 111–12, 115–17; Ex. 1007, Abstr., ¶¶ 9, 11, 15–16, 57, 65–66; Ex. 1009, 2:32–3:8; Ex. 1010, 12, 59; Ex. 1022, 184; Ex. 1023, 149; Ex. 2020, 148:14–22, 149:18–150:20, 169:13–17; Ex. 2044 ¶¶ 58–79, 81–87, 90–91, 96, 100–105, 119–128, 131–159; Ex. 2047; Ex. 2048; Ex. 2049 (“Tsuji”)). Part of Patent Owner’s opposition consists of its allegations that evidence of objective indicia of non-obviousness weighs against Petitioner’s case for obviousness. We have addressed this above as unpersuasive (*see supra* Section II.E).

Having considered the arguments and evidence presented by the parties, we determine that Petitioner has shown that the challenged claims would have been unpatentable as obvious over the cited references. We review the parties’ positions and provide our analysis below.

1. Petitioner’s Positions

Petitioner asserts that: (1) Lenhard discloses a non-invasive CRC screening method where a fecal sample is collected and separated into two portions so that a first portion is tested for blood using gFOBT and the second portion is tested for tumor-derived DNA (i.e., nucleic acids) using methylation of the HIC1 gene promoter as a detectable biomarker, and that the combination of the two tests was advantageous in increasing detection of CRCs; (2) Vilkin discloses an improved fecal-blood-detection technique, that is, iFOBT paired with buffer-stabilization of the sample’s Hb (suggested to be collected at home, and separated from a larger sample), which would have been an obvious replacement for and improvement on Lenhard’s gFOBT because it was more sensitive, eliminated diet restrictions, and provided quantitative results; and (3) Itzkowitz discloses that adding a DNA

stabilizing buffer to a fecal sample to be used for CRC screening (collected at home) improves the technique and would be an obvious addition to Lenhard's process and would allow shipment without freezing the sample. Pet. 19–24 (citing Ex. 1002 ¶¶ 145–168; Ex. 1004, 142–45, 147; Ex. 1005, 2519–20, 2524; Ex. 1006, 111–12, 116–17; Ex. 1020, 29–30; Ex. 1021, 214; Ex. 1022, 186, 190).

Petitioner asserts that the ordinarily skilled artisan would have recognized from Lenhard, Vilkin, and Itzkowitz that a non-invasive, home-collection process, where stool samples were portioned and stabilized for shipment, and then tested for the purpose of CRC screening by assaying for blood using iFOBT and assaying for DNA markers, would have obvious benefits to an ordinarily skilled artisan (such as, that it would result in “increased detection rates” and “give colorectal cancer screening a new future,” for example). *Id.* Petitioner asserts that Vilkin and Itzkowitz's teachings would have been combined with Lenhard's to improve Lenhard's techniques, as explained above (because it would provide fewer restrictions on patients, better results, etc.), with reasonable expectation of success, because the references themselves indicate that their techniques were complementary and worked, as did several other contemporary references. *Id.* (citing, *inter alia*, Ex. 1002 ¶¶ 160–167; Exs. 1011–1013, 1020–1022).

Petitioner also adds certain teachings of Kanaoka to this combination of Lenhard, Vilkin, and Itzkowitz for Kanaoka's disclosure of measuring RNA (like Lenhard's and Itzkowitz's DNA) as another, similar way to screen for CRC. Pet. 35–37 (citing Ex. 1002 ¶¶ 220–226; Ex. 1007 ¶¶ 21–26, 47–48, 55–68). Petitioner asserts that Kanaoka teaches that using RNA markers, such as COX-2, provides high detection sensitivity (for CRC),

which would further complement Lenhard's techniques (as well as the iFOBT taught by Vilkin) and would improve the screening with a reasonable expectation of success. *Id.* (further citing Ex. 1024, 131 (identifying a commercially available DNA and RNA preservative)).

After addressing the reasons the ordinarily skilled artisan would have had for combining the asserted prior art, Petitioner maps the disclosures of the references to the limitations and elements of the challenged claims 3, 4, and 12–19.

Beginning with claim 3, which incorporates the limitations of disclaimed independent claim 1 from which it depends, Petitioner asserts that Lenhard discloses the general subject matter of the claims, including (as part of a CRC screening procedure) collecting and separating into two portions a human fecal sample, and testing a first portion for extracted tumor-derived DNA (nucleic acids) and a second portion for blood proteins. Pet. 24–31 (citing, *inter alia*, Ex. 1002 ¶¶ 170–190, 195–207; Ex. 1004 (in its entirety)). Petitioner asserts that each of Vilkin and Itzkowitz teaches not freezing the samples, as well as collecting the samples at home (before shipping to a lab). *Id.* at 24–25 (citing Ex. 1002 ¶¶ 170–178; Ex. 1005, 2020; Ex. 1006, 112). Petitioner asserts that both Lenhard and Itzkowitz teach sealing the first portion of the sample in a container for subsequent DNA testing, and that Itzkowitz teaches adding a respective stabilizing buffer solution to preserve the sample, which is then tested at a lab by extracting DNA and assaying for methylated DNA indicative of CRC. *Id.* at 25–26, 29–30 (citing Ex. 1002 ¶¶ 179–186, 199–202; Ex. 1004, 143–45; Ex. 1006, 111–12, 116). Petitioner asserts that Lenhard and Vilkin each teaches sealing the second portion of the sample in a container for

subsequent blood testing, and that Vilkin teaches providing an amount of Hb stabilizing buffer to preserve the sample, which is then quantitatively tested at a lab for human hemoglobin proteins. *Id.* at 26–28 (citing Ex. 1002 ¶¶ 169–190, 194–207; Ex. 1004, 143–45; Ex. 1005, 2519–22).

Turning to claim 4, which depends from claim 3 and adds “wherein testing the nucleic acid comprises determining expression from a human gene,” Petitioner asserts this is taught by Lenhard in teaching testing a sample for methylated human DNA, which identifies expression from a human gene, and is also taught by Itzkowitz disclosing testing for methylation of the vimentin gene. *Id.* at 31–32 (citing Ex. 1001, 12:48–52 (describing epigenetic modification as any alteration in DNA, e.g., resulting in diminished gene expression); Ex. 1002 ¶¶ 208–211; Ex. 1004, 144; Ex. 1006, 111).

As for claim 12, which depends from claim 4 and adds “wherein determining expression from the human gene comprises measuring an amount of RNA expressed from the human gene,” Petitioner asserts that this would have been obvious because Kanaoka teaches that COX-2 RNA is a biomarker for CRC screening providing high sensitivity and 100% specificity, that Kanaoka teaches such an RNA assay combined with iFOBT testing, and that there were known stabilizing buffers (e.g., RNeasy) for preserving DNA and RNA at room temperature for days, which would have made it obvious to add Kanaoka’s teachings to those of Lenhard, Vilkin, and Itzkowitz. *Id.* at 35–37 (citing Ex. 1002 ¶¶ 220–224; Ex. 1007 ¶¶ 21–26, 55–68, 76; Ex. 1024, 131).

Claim 13 depends from claim 12 and adds “wherein measuring an amount of RNA expressed from the gene comprises reverse transcriptase

polymerase chain reaction (RT-PCR),” which Petitioner asserts is taught by Kanaoka disclosing employing RT-PCR in amplifying RNA for its assay. *Id.* at 37 (citing Ex. 1002 ¶¶ 225–226; Ex. 1007 ¶¶ 21–25, 47–48, 59).

Claims 14 (depending from claim 3) and 15 (depending from claim 14) respectively, add “wherein testing for an amount of blood protein present in the second portion comprises testing for a concentration of hemoglobin in the second portion, wherein a concentration of hemoglobin is indicative of a presence of blood in the fecal sample” and “wherein the testing for the concentration of hemoglobin comprises immunochemical detection of hemoglobin,” which Petitioner asserts is taught by Vilkin because it teaches using iFOBT (which would replace Lenhard’s gFOBT because it is superior thereto), where the “i” in iFOBT indicates it is an immunochemical process that quantitatively detects hemoglobin concentration. *Id.* at 32–33 (citing Ex. 1002 ¶¶ 212–215; Ex. 1005, 2519–22).

Finally, as for claims 16–19, which depend from claim 14, each adds a threshold hemoglobin concentration indicating a positive result of screening, i.e., 5, 10, 20, and 50 ng/ml, and Petitioner asserts that these would have been obvious over the prior art combination because hemoglobin concentration detected is a result-effective variable that would have been routinely optimized by the ordinarily skilled artisan. *Id.* at 33–34 (citing Ex. 1002 ¶¶ 216–219). Petitioner asserts, “Vilkin explains that one of the advantages of its test is that it ‘allows the physician to choose the optimal fecal Hb threshold level that triggers a follow-up colonoscopy’” and “[s]uch threshold selections ‘involve an evaluation of cost-benefit.’” *Id.* (citing Ex. 1005, 2520–24; Ex. 2021, 210; Ex. 1023, 140).

Having summarized Petitioner's positions on the prior art combinations and their teaching of the claimed subject matter, we turn to Patent Owner's positions in opposition.

2. Patent Owner's Positions

We have already addressed Patent Owner's arguments relating to objective indicia of non-obviousness above, concluded that they are entitled to little weight. We review Patent Owner's other arguments below.

Patent Owner takes the over-arching position (as to all of Petitioner's Grounds for unpatentability) that there was no motivation, based on the asserted prior art combination, to perform the claimed at-home separation of stool sample portions into two separate containers with respective stabilizing buffer/solution (before analyzing those samples). Resp. 21–24 (citing Ex. 1006, 1; Ex. 1008, 4; Ex. 1022, 2; Ex. 2020, 148:14–22, 149:18–150:20; Ex. 2044 ¶¶ 119–128; Ex. 2048; Ex. 2049). Patent Owner's rationale for this argument is that, at the time of the invention (i.e., 2009), the person of ordinary skill in the art would have understood or believed that a single stabilizing buffer, such as a Tris-EDTA solution disclosed or taught by Itzkowitz (via citation to Olson, Ex. 1022) and Shuber, and also disclosed in Tsuji (Ex. 2049, 10–11), or a commercially available product called “RNAlater,” would be sufficient to stabilize and preserve both nucleic acid (e.g., DNA) and blood protein (e.g., hemoglobin). Resp. 21–24. Patent Owner's point is that, if a single buffer solution would have preserved both the CRC screening targets of DNA and human blood hemoglobin, “[a]bsent a compelling reason,” a patient or the person of ordinary skill in the art would not choose a system requiring the at-home manipulation of a stool sample and the use of two buffer solutions. *Id.* (citing Ex. 2044 ¶ 128).

Patent Owner also argues that only hindsight would have allowed the ordinarily skilled artisan to combine Lenhard, Vilkin, and Itzkowitz. Resp. 24–26. Patent Owner argues that it would not have been obvious to change Lenhard’s method (in what Patent Owner calls “multiple significant modifications”), which included freezing samples, so as to use un-frozen samples, or to do sample collection at home, or to use Vilkin’s stool probe/container, or to perform *both* blood and nucleic acid analyses. *Id.* Patent Owner cites Vilkin’s method including collecting three stool samples over 3 days, and the required home-refrigeration of samples before shipping to a lab, as issues that would make patients hesitant to adopt the technique. *Id.* at 24–26, 33–34 (citing, *inter alia*, Ex. 1005, 2519–20, 2524; Ex. 2044 ¶¶ 83–87, 148–151). Patent Owner argues that Itzkowitz’s methods (which are acknowledged as providing a “sealable . . . collection kit” for patients) do not require sample portioning, which was preferred by patients. *Id.* at 26.

Patent Owner argues that Lenhard suggests using one, not multiple classes of markers, and did not provide statistically significant results to persuade an ordinarily skilled artisan to pursue its methods. *Id.* at 27–29 (citing, *inter alia*, Ex. 2044 ¶¶ 58–79, 131–146). Patent Owner also argues that the results of Vilkin and Itzkowitz did not produce sufficient sensitivity and specificity in screening CRC to persuade an ordinarily skilled artisan to adopt their methods (although, Patent Owner identifies that Vilkin reports “all cases of cancer and most advanced adenomas would be detected and a negative test would provide a very high degree of certainty,” and calls Itzkowitz’s DNA assay an “optimal combination [that] detected most cancers”). *Id.* at 30–31. And, Patent Owner calls the endeavor to use multiple markers for CRC screening “highly unpredictable.” *Id.* at 31.

Patent Owner argues that the prior art's disclosed reduction in specificity would have actually discouraged its combination. *Id.* at 31–32. Finally, Patent Owner argues that a method based on combining Lenhard, Vilkin, and Itzkowitz would add complexity, which would not have motivated an ordinarily skilled artisan to pursue the combination. *Id.* at 35–36 (citing, *inter alia*, Ex. 2044 ¶¶ 147–159).

Regarding the combination of Lenhard, Vilkin, Itzkowitz, and Kanaoka, Patent Owner argues that Kanaoka requires freezing of the sample, which is expressly not a part of the claimed invention. Resp. 47, 49 (citing Ex. 1007 ¶¶ 57, 65; Ex. 2044 ¶¶ 98–100, 179–193). Patent Owner also argues that, according to Kanaoka, combining iFOBT testing and RNA testing results in only marginally improved results (one additional positive test), and potentially some erroneous tests (false positives with iFOBT), which would not have motivated the ordinarily skilled artisan to add Kanaoka to the prior art combination. *Id.* at 48 (citing Ex. 1007 ¶¶ 65–67; Ex. 2044 ¶¶ 183–184). And, Patent Owner argues that Kanaoka suggests RNA is too unstable and difficult to preserve to have been a good candidate for CRC screening based on an at-home sample collection. *Id.* at 49–50 (citing Ex. 1007 ¶¶ 8–9, 11; Ex. 1024, 125, 131 (Patent Owner discounts Nechvatal's disclosure of RNA preservative and screening potential); Ex. 2044 ¶¶ 188–191, 199). Finally, Patent Owner argues that Nechtaval (Ex. 1024) reports that the product RNAlater can preserve both RNA and protein (e.g., hemoglobin); thus, if Kanaoka were considered, an ordinarily skilled artisan would have seen no reason for the sample-portioning of the claims. *Id.* at 51 (citing, *inter alia*, Ex. 2044 ¶¶ 175–193).

Below, we analyze these arguments in the context of the record.

3. Analysis

As explained further below, we conclude that Petitioner has shown that the combination of Lenhard, Vilkin, and Itzkowitz teaches or suggests the invention of challenged claims 3, 4, and 14–19, and that this prior art combination, with the addition of Kanaoka, teaches and suggests the invention of claims 12 and 13. Furthermore, we agree with Petitioner that the person of ordinary skill in the art would have combined these references’ teachings with a reasonable expectation of success. We adopt Petitioner’s arguments and evidence as to claims 3, 4, and 12–19 under Grounds 1 and 2. *See supra* Section II.F.1 (review of Petitioner’s assertions regarding dependent claims). We are not persuaded by Patent Owner’s arguments in opposition, which we also address below.

It is worth first noting that the ’746 patent states that, as of its invention, CRC screening using iFOBT tests and DNA tests (including DNA methylation) was not new—these techniques were known, individually, as alternatives to colonoscopy. Ex. 1001, 1:56–3:22 (the ’746 patent’s Background Section calling fecal occult blood testing “commonly used” and DNA testing a “[r]ecent development[],” but identifying more than one commercial product available for such tests). The ’746 patent, however, foundationally describes its invention as CRC screening by testing for *both* the presence of blood and for an epigenetic DNA modification in a fecal sample (as a non-invasive alternative to colonoscopy). *Id.* at 4:3–16 (Summary of the Invention), 6:50–7:12; *see also id.* at claims 1 and 3. And, the ’746 patent foundationally describes that this combination of tests is an innovation because, “[a]lthough combined assays for detecting CRC ha[d] been described, their approach targets either multiple protein markers or

either multiple DNA alterations,” and “[t]o date, immunochemical tests and DNA tests for CRC detection have been evaluated and compared on a separate basis only.” *Id.* at 3:41–45. However, on this record, it is apparent that this statement in the ’746 patent is not entirely correct.

Combined blood tests and DNA tests for screening for CRC were known, as these two types of CRC screening tests (blood and DNA) were disclosed as performed on the same fecal samples, and FOBT and DNA (or RNA) tests were disclosed in the prior art as complementary to one another (or at least suggested to be so) for CRC screening, several times. *See, e.g.*, Ex. 1004; Ex. 1007 ¶¶ 55–68 (Example 1); Ex. 1011, 112; Ex. 1012, 33 (Abstr.); Ex. 1013, 1571, 1578; Ex. 1043; *see also* Ex. 1002 ¶¶ 33 n.6, 50 n.9, 63–66, 100 (identifying the existence of combined DNA analysis and occult blood testing as of priority date). This, alone, does not prove Petitioner’s challenges, but it does provide context for Petitioner’s arguments and evidence, and for the alleged invention.

Petitioner’s evidence shows that, before the priority date of the challenged claims, CRC screening by processing two portions of a human fecal sample to detect the presence of blood and epigenetically modified DNA (isolated, bisulfite-treated, purified, and PCR-amplified) was taught by Lenhard, which disclosed that a combined gFOBT+hypermethylated HIC1 detection method worked well to detect CRCs; and “[t]he combination of both assays resulted in increased detection rates for CRCs.” Ex. 1004, 142–43, 145–48; *see also* Ex. 1002 ¶¶ 113–119, 144–148, 150, 153–155, 158, 163–165, 171–172, 174, 179, 187, 200–201, 210 (discussing Lenhard). Other possible genetic markers for CRC were also identified by Lenhard; e.g., hypermethylation at SFRP2, SFRP5, PGR, hMLH1, and HPP1, or other

methylation markers, possibly used in combination with one another. Ex. 1004, 143, 148. Thus, Lenhard teaches a fecal screening test for CRCs utilizing *combined* blood and nucleic acid assays on a respectively dual-portioned sample, but Lenhard is not the only incidence or suggestion of such in the prior art of record.

In 2001, Nishikawa disclosed a CRC screening method using DNA extracted from stool to detect mutations that may identify cancer, compared it to FOBT screening of the sample, and concluded that the DNA testing was “highly specific, noninvasive, and cost-effective, and ***should provide a more sensitive and specific tool for mass screening of colorectal cancer*** than is currently available, ***especially if used in combination with fecal occult blood testing.***” Ex. 1011, 107–08, 112 (emphasis added). Thus, Nishikawa teaches the combined assay technique.

In 2005, Kutzner disclosed that a combined FOBT and genetic screen (called “molecular diagnosis (MD)”) for target DNA alterations was effective to screen for CRC and “[t]he ***combined application*** of the two methods together led to a ***significantly higher sensitivity*** than the application of any of the methods alone,” and “suggest[ed] that ***MD and FOBT should complement rather than replace each other.***” Ex. 1012, 33–37, 40 (emphasis added). Thus, Kutzner teaches the combined technique.

And, in 2008, Levin disclosed that the “range of options for CRC screening” included “2 general categories: stool tests, which include tests for occult blood or exfoliated DNA,” such as guaiac-based fecal occult blood tests and fecal immunological tests (respectively, gFOBT and FIT), and structural exams, which include, *inter alia*, “colonoscopy (CSPY),” and further disclosed that “[t]hese ***tests may be used alone or in combination,***”

that “[c]ollection kits [for FOBT and DNA testing] ha[d] been designed to facilitate specimen collection and mailing and to enhance compliance,” and reported that data showed “***combined results from a standard gFOBT and a panel of DNA markers*** (APC, BAT26, and L-DNA) resulted in a ***combined sensitivity for cancer of 93% and specificity of 89%***.” Ex. 1013, 1570–78 (emphasis added). Thus, Levin teaches the combined assay technique.

None of Nishikawa, Levin, or Kutzner is directly asserted here by Petitioner against the claims as prior art, but each supports Petitioner’s position, as explained by Petitioner’s declarant, that, before the priority date of the ’746 patent, there was nothing new or non-obvious about screening for CRC by detecting both blood and DNA markers in a respectively portioned fecal sample, as taught by Lenhard. See Ex. 1002 ¶¶ 33, 50, 54, 61, 63–66, 100, 233, 235–236 (discussing Nishikawa, Levin, and Kutzner).

As noted above, Lenhard disclosed the detection of blood in fecal samples using gFOBT (a blue color reaction indicated a positive result) in a portion of a fecal sample separated from another portion. Ex. 1004. Vilkin, which is also focused on non-invasive CRC screening, teaches that, as of 2005, there was a better test for blood in fecal samples—iFOBT (immunochemical detection of Hb). Ex. 1005, 2519.

Vilkin discloses that it was known that gFOBT had “low sensitivity for . . . CRC . . . and low specificity due to nonspecificity for human hemoglobin”; it required “diet restrictions” as well. *Id.* Vilkin explains that the iFOBT screen overcame these drawbacks and also offered quantitative results whereby a threshold could be set for colonoscopy. *Id.* Moreover, of some relevance in this proceeding, in particular, Vilkin discloses that its iFOBT screening utilized a commercially available product and system

composed of a sample probe and storage tube (with an Hb stabilizing buffer) for patient sample collection and shipping to a lab (unfrozen), and an automated counter-top iFOBT analysis instrument called the Eiken “OC-Sensor,” already of use in Japan, Italy, and Uruguay (other automated systems, e.g., “InSure” and “Magstream,” were used, at least investigatively, in the United States, Australia, Japan, France and/or Hong Kong). *Id.* at 2520, 2524–25.

Petitioner combines Vilkin’s teachings with Lenhard’s for this subject matter, substituting Lenhard’s gFOBT screen with Vilkin’s better iFOBT screen, and with it Vilkin’s associated sampling device (tube and probe) for portioning (at the patient’s home), storing, and preserving in an Hb stabilizing buffer, the fecal sample portion to be so tested, which did not require freezing, but only needed refrigeration until its return to a lab for testing. *Id.* at 2520. With such a modification to Lenhard’s methods and materials, much of the required steps of claim 3 are taught and obvious (i.e., subject matter of claim 1 requiring at-home sample collection, portioning, use of a blood protein stabilizer, and a sealed vessel; and claim 3’s testing for blood protein using that sample).

Petitioner’s position on why an ordinarily skilled artisan would have combined Vilkin with Lenhard is persuasive.

First, the two references are focused on (noninvasive) CRC screening and use a stool-blood-assay to do it, so they are from the same field of endeavor and pertinent to the problems of the ’746 patent’s invention; thus, they are analogous art and properly combinable to establish obviousness. *In re Bigio*, 381 F.3d 1320, 1325 (Fed. Cir. 2004) (“Two separate tests define the scope of analogous prior art: (1) whether the art is from the same field of

endeavor, regardless of the problem addressed and, (2) if the reference is not within the field of the inventor's endeavor, whether the reference still is reasonably pertinent to the particular problem with which the inventor is involved.”).

Moreover, Vilkin identifies Lenhard’s gFOBT as inferior to its own iFOBT because: (1) iFOBT was specific for human Hb; (2) had sufficient (acceptable) specificity, better and quantitative sensitivity; (3) used a convenient, available, and self-contained sampling device (with included Hb stabilizing buffer); (4) did not require sample freezing (used refrigeration) and the sample was stable for weeks; (5) used a known and commercially available automated desktop analyzer; (6) samples could be collected at home; and (7) iFOBT overcame the need for dietary restrictions of gFOBT. *See generally* Ex. 1005. We find these to be very good reasons, with rational underpinning, why the ordinarily skilled artisan would have modified Lenhard’s respective methods and materials to use those of Vilkin, with a reasonable expectation of success.¹⁹ *See, e.g.*, Ex. 1002 ¶¶ 144–158, 165–168 (discussing reasons to combine Vilkin and Lenhard).

As noted above, Lenhard teaches screening of epigenetically modified nucleic acids (i.e., a DNA assay), but does not mention sample-preserving with a stabilizing buffer. For this, Petitioner points to the teachings of Itzkowitz, which discloses that the immediate addition of a DNA preservation buffer to a fecal sample is advantageous because it prevents sample degradation for several days, even at room temperature, and allows

¹⁹ We note that Vilkin also expresses a need for an improved and more convenient, at-home, toilet-based stool-collecting and sample storage device or system to improve upon a zip-lock bag system. Ex. 1005, 2524. As discussed below, Itzkowitz meets this need.

higher screening sensitivity—better for detecting cancer—and the test had “a very high degree of patient satisfaction.” Ex. 1006, 111–17. Moreover, the patients providing samples in Itzkowitz’s procedures were given “a special stool collection kit for mounting to a toilet,” where the buffer was immediately added to the collected sample, and then the samples were “shipped at room temperature overnight” to a clinical lab. *Id.* at 112. Similar to Vilkin, Itzkowitz’s procedures suggest collecting specimens at home directly in the same sealable container in which the sample (one portion) is shipped.

By modifying Lenhard’s methods and materials to add Itzkowitz’s nucleic acid preserving buffer, and collection materials and techniques, all steps and elements of claim 1 are addressed in the prior art. Moreover, simply performing the intended assays, using standard techniques, as disclosed by these references, teaches the steps and elements of claim 3.

And, like Vilkin, Itzkowitz is from the same field of endeavor (noninvasive CRC screening) as Lenhard (and the ’746 patent) and addresses problems pertinent to the invention of the ’746 patent, namely CRC screening using DNA analysis, thus the references are analogous art. *In re Bigio*, 381 F.3d at 1325. Moreover, the prospect of convenient at-home sample collection, predictable sample preservation using a known buffer, and improved results, as taught and suggested by Itzkowitz, are good reasons (with rational underpinning) for the ordinarily skilled artisan to have combined Itzkowitz with Lenhard (and Vilkin), with a reasonable expectation of success. *See* Ex. 1002 ¶¶ 144, 154–168 (discussing reasons to combine Itzkowitz with Lenhard).

Turning to Kanaoka, like Vilkin and Itzkowitz, it is also focused on CRC screening using non-invasive testing (assaying for nucleic acids in the form of RNA rather than DNA). Thus, it is also analogous art. *In re Bigio*, 381 F.3d at 1325. Further, Kanaoka teaches that the use of RNA markers, such as COX-2 RNA, is another useful way in addition to DNA assays and FOBT to detect CRC, which potentially provides higher detection sensitivity and specificity, while requiring smaller samples. Ex. 1007 ¶¶ 2–8, 21, 65–66, 76. Thus, we agree with Petitioner that the ordinarily skilled artisan would have had a reason to combine Kanaoka with Lenhard, Vilkin, and Itzkowitz, with a reasonable expectation of success. Ex. 1002 ¶¶ 220–223 (discussing reasons for adding Kanaoka to the prior art combination).

Having found that the record strongly supports that the asserted prior art would have been combined by the ordinarily skilled artisan, as asserted by Petitioner, we also find that the combination of Lenhard, Vilkin, and Itzkowitz teaches or suggests each step and element of claims 3, 4, and 14–19, and that the addition to this prior art combination of the subject matter taught by Kanaoka teaches or suggests the subject matter of claims 12 and 13. *See* Pet. 19–37; Ex. 1002 ¶¶ 169–226. We agree with and adopt Petitioner’s reasoning and assertions of fact on this as our own. *See supra* Section II.F.1 and above-cited portions of the Petition.

Petitioner accounts for all claim elements of the challenged claims as taught or suggested by the prior art combination(s). To summarize, as asserted by Petitioner, in combination, Lenhard, Vilkin, and Itzkowitz teach a method of (and the materials for) CRC screening where a patient collects a fecal sample at home using a special stool collection kit mounted to a toilet, divides that sample into a first portion for iFOBT assay using a storage tube

and probe device and a second portion for DNA marker assay, sealing each portion in a respective vessel with a stabilizing buffer/solution, and shipping the sample to a lab, all without freezing the sample. *See* Exs. 1004–1006. Then, the lab, using respective sample portions, conducts a quantitative iFOBT test to detect an Hb concentration and extracts DNA and assays for epigenetic markers (e.g., methylated genes/promoters) indicative of cancer—this is the method of claims 3, 4, 14, and 15. *Id.* And, as asserted by Petitioner, the thresholds for Hb concentrations recited by claims 16–19 are result-effective variables and would have been routinely optimizable by and obvious to an ordinarily skilled artisan. Finally, adding Kanaoka to the prior art combination, as asserted by Petitioner, adds a COX-2 RNA assay to the CRC screening, where the RNA is provided for assay using, *inter alia*, RT-PCR amplification—this is the method of claims 12 and 13.

As noted above in our summary of Patent Owner’s positions, Patent Owner does not argue that any step or element of the claims is missing from the combined prior art, but argues that the combination(s) would not have been made, that Petitioner’s asserted way of combining the prior art to achieve the invention requires improper hindsight, and that the invention was non-obvious because there was a more preferred way of preserving a fecal sample.

We have already discussed above that Patent Owner’s arguments over objective indicia of non-obviousness are not persuasive. *See supra* Section II.E. We now address Patent Owner’s other arguments.

Contrary to Patent Owner’s arguments, we find no unreasonable “picking and choosing” in Petitioner’s citations to the teachings of Vilkin and Itzkowitz because the subject matter they disclose, and that would be

incorporated into the methods of Lenhard, is straightforward and is a substantial focus of each reference rather than a mere component that might be selected from a number of options. *See* Ex. 1005; Ex. 1006. Therefore, we move on to Patent Owner's other arguments.

Patent Owner's argument that the ordinarily skilled artisan would have looked to a single preservative buffer for a fecal sample and, thus, would not have been steered toward the at-home portioning and respective stabilizing solutions/buffers of the claims (or of a combination of Lenhard, Vilkin, and Itzkowitz), in the abstract, makes sense. However, upon consideration of both parties' arguments and the evidence of record, we are not persuaded by Patent Owner's position.

Based on the record, it appears that it *may* have been possible to use a single buffer solution to preserve both hemoglobin and DNA for shipping a fecal sample without freezing it—that buffer potentially being a solution of Tris-EDTA (and likely also a salt) or a commercial product called RNAlater. This is supported by the disclosure of Itzkowitz, which cites Olson, which teaches that “a DNA-stabilizing buffer” can be “0.5 mol/L Tris, 0.15 mol/L EDTA and 10 mmol/L NaCl.” Ex. 1006, 112, 117 (citing Olson); Ex. 1022, 184. It is also supported by the disclosure of Tsuji, which teaches that a buffer solution for human hemoglobin containing fecal matter can include, *inter alia*, “Tris-HCl buffer,” and may also include a salt and EDTA. Ex. 2049 ¶¶ 9–11. This is further supported by the disclosure of the product RNAlater, which appears to be a commercial product for preserving both RNA (nucleic acids) and proteins (hemoglobin is a protein). Ex. 2048. This evidence supports that a single-preservative-buffer solution may have been possible and obvious for the asserted purpose. However, the obviousness of

a single-preservative-buffer solution does not mean that the two-portion and respective preservative-buffers taught by the asserted prior art was not also obvious.

In analyzing whether there existed a reason to combine prior art, as asserted here, we must consider “the so-called ‘known technique’ rationale,” where “if there’s a known technique to address a known problem using ‘prior art elements according to their established functions,’ then there is a motivation to combine,” because *KSR* explains that “‘if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill.’” *Intel*, 61 F.4th at 1380 (quoting *KSR*, 550 U.S. at 417; and *Intel*, 21 F.4th at).

It is clear on this record that what could be called “off the shelf” techniques and materials taught by Vilkin and Itzkowitz were techniques and materials used to improve non-invasive CRC screening and that they were, based on their and Lenhard’s teachings, readily available and within the skill or an ordinarily skilled artisan to combine them.

If addressing a known problem, like the need to improve non-invasive CRC screening in these circumstances, the prior art combination need not be “the *best* option, only . . . a *suitable* option.” *Id.* (quoting *Intel*, 21 F.4th at 800 (emphasis original)). Simply using the available techniques and materials of Lenhard, Vilkin, and Itzkowitz, as the references straightforwardly instruct, would have been a suitable option for the person of ordinary skill in the art to improve non-invasive CRC screening (e.g., an alternative or supplement to colonoscopy). Perhaps it would not have been the “best” option because, potentially, a technique could have been

developed where the patient need not manipulate their fecal sample to portion it, but the law of obviousness does not require an option to be the best option to be obvious. For these reasons, bearing in mind that the ultimate burden here is always on Petitioner, we are not persuaded by Patent Owner's argument.

We also reject Patent Owner's argument that only impermissible hindsight would have led the ordinarily skilled artisan to this asserted prior art combination. Of course, "[i]t is impermissible . . . simply to engage in a hindsight reconstruction of the claimed invention, using the [challenged patent's] structure as a template and selecting elements from references to fill the gaps." *In re Gorman*, 933 F.2d 982, 987 (Fed. Cir. 1991). But, where a skilled artisan would pursue "known options" from "a finite number of identified, predictable solutions," the resulting invention is obvious. *KSR*, 550 U.S. at 421. "Any judgment on obviousness is in a sense necessarily a reconstruction based upon hindsight reasoning, but so long as it takes into account only knowledge which was within the level of ordinary skill at the time the claimed invention was made and does not include knowledge gleaned only from applicant's disclosure, such a reconstruction is proper." *In re McLaughlin*, 443 F.2d 1392, 1395 (CCPA 1971). Such is the case here because Lenhard, Vilkin, and Itzkowitz (and Kanaoka) disclose the claimed steps and elements to have been advantageously used for the same purposes as the invention; no improper hindsight reconstruction was required.

As to Patent Owner's arguments regarding the references teaching using one versus several markers for CRC, whether the references' established improvements warranting their adoption and combination, the

alleged unpredictability of the field, and the skilled artisan's desire not to add complexity to existing techniques, we find them unavailing.

Lenhard indicated its detection method was “sensitive and specific” and “well suited for stool-based CRC screening” and that adding more markers could “allow for the highly sensitive and specific stool-based detection of CRCs and adenomas.” Ex. 1004, 148. Moreover, Vilkin teaches that, by using its improved iFOBT method, one of ordinary skill can achieve a high sensitivity of 76.5% and an acceptable specificity of 95.3%. Ex. 1005, 2523. Finally, Itzkowitz explains that the improvements in sensitivity offered by its methods came “at the expense of a relatively minor decrease in specificity.” Ex. 1006, 116. The prior art here is clear as to the advantages, capabilities (and improvements), and expected results of the respective techniques. The prior art asserted here also addresses Patent Owner's identified concerns as acceptable minor detractors from the advantages of their methods. *See Medichem S.A. v. Rolabo S.L.*, 437 F.3d 1157, 1165 (Fed. Cir. 2006) (“The fact that the motivating benefit comes at the expense of another benefit, however, should not nullify its use as a basis to modify the disclosure of one reference with the teachings of another. Instead, the benefits, both lost and gained, should be weighed against one another.”). Furthermore, “[t]hat a given combination would not be made by businessmen for economic reasons does not mean that persons skilled in the art would not make the combination because of some technological incompatibility. Only the latter fact would be relevant.” *In re Farrenkopf*, 713 F.2d 714, 718 (Fed. Cir. 1983) (citing *Orthopedic Equip. Co. v. United States*, 702 F.2d 1005, 1013 (Fed. Cir. 1983)).

As for Patent Owner’s arguments concerning Kanaoka, e.g., that it requires freezing of a sample and that the results were not good enough to entice its combination with the other prior art, we are not persuaded.

First, Kanaoka is explicit that, in its methods, “biological samples . . . can be used as they are, or, in some cases, after frozen.” Ex. 1007 ¶ 41. Freezing may be useful, but it is not required—in this context, “as they are” means unfrozen. And, adding another biomarker to the Lenhard-Vilkin-Itzkowitz combined teachings,²⁰ specifically COX-2 RNA, would have been attractive to the ordinarily skilled artisan because Kanaoka discloses that its process works (high sensitivity and specificity) and also that a COX-2 assay could be effectively combined with iFOBT in a complementary way (each detecting certain CRC the other may not). Ex. 1002 ¶¶ 220–223; Ex. 1007 ¶¶ 55–68. Moreover, Kanaoka states that “the detection of *COX-2* from the feces by the RT-PCR technique had 90% sensitivity and 100% specificity, and it is proved that the present invention is superior to a conventional technique of the immunological fecal occult blood test,” which suggests using such a technique with the methods of the other asserted prior art. Ex. 1007 ¶ 75. Thus, the technique worked and had advantages, which supports Petitioner’s allegation that one of skill in the art would have been motivated to combine Kanaoka’s teachings with the other asserted prior art.

4. Summary and Conclusions

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 3,

²⁰ Lenhard (Ex. 1004, 143, 147–48) suggests using other biomarkers (e.g., SFRP2, SFRP5, PGR were known) in addition to methylated HIC1, and Itzkowitz (Ex. 1006) teaches targeting two genetic biomarkers—e.g., “DY” and vimentin.

4, and 14–19 are unpatentable as obvious over Lenhard, Vilkin, and Itzkowitz, and that claims 12 and 13 are unpatentable as obvious over this same prior art combination adding Kanaoka.

G. OBVIOUSNESS OVER SHUBER AND VILKIN, AND ALSO COMBINED WITH KANAOKA (GROUNDS 3 AND 4)

We conclude above that all claims have been proven unpatentable by a preponderance of the evidence under Grounds 1 and 2, but for completeness, we review and analyze Petitioner’s Grounds 3 and 4 below.

Petitioner contends, under Ground 3, that claims 3, 4, and 14–19 would have been obvious over Shuber and Vilkin. Pet. 12, 37–48 (citing Ex. 1002 ¶¶ 227–271; Ex. 1004, 143, 147; Ex. 1005; Ex. 1008; Ex. 1011, 112; Ex. 1012, 34; Ex. 1013, 1578; Ex. 1033, 20–33). Petitioner also asserts, under Ground 4, that claims 12 and 13 would have been obvious over Shuber, Vilkin, and Kanaoka. Pet. 39–49 (citing, *inter alia*, Ex. 1002 ¶¶ 272–277; Ex. 1007 ¶¶ 21–25, 47–49, 59). The teachings of Vilkin and Kanaoka are cited and asserted by Petitioner here similarly to Grounds 1 and 2. *See supra* Section II.F.1; *see also supra* Section II.D (discussing these references).

Patent Owner opposes Petitioner’s challenges. Resp. 21–24, 37–45, 47–51 (citing, *inter alia*, Ex. 1001, 45:59–61, 45:63–67, 46:56–57; Ex. 1003, 3; Ex. 1004; Ex. 1005; Ex. 1008; Ex. 1011; Ex. 1012; Ex. 1013; Ex. 2044 ¶¶ 58–87, 109, 131–140, 147–159, 160–176). Some of Patent Owner’s opposition to Petitioner’s Grounds 3 and 4 is generic to all Grounds, has been addressed above as not sufficient to overcome Petitioner’s strong showing of obviousness, and will not be revisited in detail here.

Having considered the arguments and evidence presented by the parties, we determine that Petitioner has shown that the challenged claims would have been unpatentable as obvious over the cited references. We explain below.

1. Petitioner's Positions

Vilkin is asserted and cited by Petitioner under Grounds 3 and 4 in much the same way it is asserted and cited under Grounds 1 and 2; that is, for teaching that iFOBT is an effective and attractive CRC screening test that can utilize essentially off-the-shelf products (the sampling probe and sealable tube with preservative buffer solution and counter-top analyzer called the OC-Sensor). Pet. 38 (citing Ex. 1005, 2520–21).

Petitioner asserts that Shuber teaches fecal DNA analysis for CRC screening, where a sample is deposited in a sealable container, with a nucleic acid stabilizing solution, and shipped to a testing center for nucleic acid assay (a hypermethylation assay), at no point requiring freezing. *Id.* at 38 (citing Ex. 1008, 7:12, 7:19–21, 10:12–14, 29:8–19). Petitioner's positions on Vilkin are already discussed above and apply here. Petitioner asserts that the person of ordinary skill in the art would have combined Shuber's and Vilkin's teachings with a reasonable expectation of success because, as of the '746 patent's priority date, it was already well understood that a fecal DNA assay and a fecal occult blood test could be used together for non-invasive CRC screening in a complementary fashion. *Id.* at 38–40 (citing Ex. 1002 ¶¶ 233–236; Ex. 1004; Exs. 1011–1013).

Based on the combined teachings of Shuber and Vilkin, Petitioner asserts that an ordinarily skilled artisan would have found it obvious to have a patient collect a fecal sample at home and separate it into two portions to

be sealed in respective vessels with respective nucleic acid (DNA) and hemoglobin stabilizing buffers/solutions for shipping to a laboratory for DNA and iFOBT assays, at no point freezing the sample, all with a reasonable expectation of success. *Id.* at 40–41 (citing Ex. 1002 ¶¶ 232–238; Ex. 1005, 2020; Ex. 1008, 10:10–15, 29:8–19).

As for adding Kanaoka to this prior art combination, Petitioner’s rationale is essentially the same as set forth under Ground 2; that is, Kanaoka teaches an additional and complementary CRC screen using RNA markers. Pet. 48–49 (citing, *inter alia*, Ex. 1002 ¶¶ 272–276); *see supra* Section II.F.1.

After addressing the reasons the ordinarily skilled artisan would have had for combining the asserted prior art, Petitioner maps the references’ disclosures to the limitations of the challenged claims 3, 4, and 12–19.

Petitioner asserts that claim 3, which depends from claim 1, would have been obvious over Shuber and Vilkin because each reference teaches using freshly-collected (not-frozen) fecal samples, and suggest doing so at home before shipping to a lab. Pet 41–42 (citing Ex. 1002 ¶¶ 240–246; Ex. 1005, 2520; Ex. 1008, 7:10–12, 29:8–19). Petitioner asserts that Shuber teaches that a portion of the sample is sealed in a container with nucleic acid stabilizing solution to later be lab-tested by DNA assay, e.g., for hypermethylation, that Vilkin teaches that a portion of such a sample is extracted and sealed in a tube with a hemoglobin stabilizing buffer to later be lab-tested by iFOBT, and that the sealed samples are shipped to a lab and then quantitatively tested in these ways. *Id.* at 42–43 (citing Ex. 1002 ¶¶ 247–250, 255–262; Ex. 1005, 2519–22; Ex. 1008, 7:19–21, 10:10–15, 15:18–16:17, 29:8–19).

Petitioner asserts that the DNA testing (detecting methylated human genes) disclosed in Shuber is an assay determining expression from a human gene, which teaches the subject matter of claim 4. Pet. 46–47 (citing Ex. 1002 ¶¶ 263–265; Ex. 1008, 14:1–5).

As for claim 12, which depends from claim 4 and adds “wherein determining expression from the human gene comprises measuring an amount of RNA expressed from the human gene,” Petitioner asserts that this would have been obvious because Kanaoka teaches that COX-2 RNA is a biomarker for CRC screening providing high sensitivity and 100% specificity, that Kanaoka teaches such an RNA assay can be combined with iFOBT testing, and that there were known stabilizing buffers (e.g., RNeasy) for preserving DNA and RNA at room temperature for days, which would have made it obvious to add Kanaoka’s teachings to those of Shuber and Vilkin. *Id.* at 48–48 (citing, *inter alia*, Ex. 1002 ¶¶ 272–276).

Claim 13 depends from claim 12 and adds “wherein measuring an amount of RNA expressed from the gene comprises reverse transcriptase polymerase chain reaction (RT-PCR),” which Petitioner asserts it taught by Kanaoka disclosing employing RT-PCR in amplifying RNA for its assay. *Id.* at 37 (citing Ex. 1002 ¶¶ 277; Ex. 1007 ¶¶ 21–25, 47–48, 59).

Claims 14 (depending from claim 3) and 15 (depending from claim 14) respectively add “wherein testing for an amount of blood protein present in the second portion comprises testing for a concentration of hemoglobin in the second portion, wherein a concentration of hemoglobin is indicative of a presence of blood in the fecal sample,” and “wherein the testing for the concentration of hemoglobin comprises immunochemical detection of hemoglobin,” which Petitioner again asserts are taught by Vilkin because the

reference teaches using quantitative iFOBT, where the “i” indicates it is an immunochemical process that quantitatively detects hemoglobin concentration. *Id.* at 47 (citing Ex. 1002 ¶¶ 266–267; Ex. 1005, 2519–22).

As for claims 16–19, which depend from claim 14 and add threshold hemoglobin concentrations that indicate a positive screening result, i.e., 5, 10, 20, and 50 ng/ml, Petitioner again asserts that these would each have been obvious over the prior art combination because detected hemoglobin concentration is a result-effective variable that would have been routinely optimized by the ordinarily skilled artisan. *Id.* at 47–48 (citing, *inter alia*, Ex. 1002 ¶¶ 268–271).

Having summarized Petitioner’s positions on the prior art combinations and their teaching of the claimed subject matter, we turn to Patent Owner’s positions in opposition.

2. Patent Owner’s Positions

We will not review Patent Owner’s over-arching arguments over a general lack of motivation to combine the prior art and objective indicia of non-obviousness, as they are already addressed above and the same conclusions apply here. They are not persuasive on this record. *See supra* Sections II.E and II.F.3.

Patent Owner argues that, without hindsight, there was no motivation for the ordinarily skilled artisan to have combined Shuber and Vilkin, pointing out that Shuber merely suggests combining different nucleic acid assays rather than combining a nucleic acid assay with some other test.²¹

²¹ Patent Owner points out that the Shuber inventors were employed by Exact Sciences and that Petitioner’s witness, Dr. Whitney, was one of these inventors. Resp. 37. Thus, Shuber appears to represent Patent Owner’s own efforts toward developing a non-invasive CRC screening test.

Resp. 37–38 (citing Ex. 1008, 28; Ex. 2044 ¶¶ 160–176). Patent Owner also argues that Shuber is silent about sample collection at home and Vilkin requires collecting three samples over three days—neither reference suggesting separating a sample into two portions. *Id.* at 39–43, 44 (citing, *inter alia*, Ex. 2044 ¶¶ 80–87, 147–151, 164–166). Based on this, Patent Owner argues that the prior art does not teach separating a stool sample at home, into two portions, sealed in respective containers, as required by the claims. *Id.* at 39.

Patent Owner argues that the ordinarily skilled artisan would have had no motivation to combine Vilkin and Shuber, arguing that adding Vilkin’s test would not improve Shuber’s test’s sensitivity, which was already ~98% for detecting adenomas. *Id.* at 39–43 (citing, *inter alia*, Ex. 1008, 14; Ex. 2044 ¶¶ 109, 153–157, 160–175). On this point, Patent Owner also argues that Nishikawa, Lenhard, Kutzner, and Levin (evidence Petitioner cites as supporting reasons the person of ordinary skill in the art would have considered combining a fecal blood and DNA test for CRC screening) are speculative and actually support the tests being used individually. *Id.* at 40– (citing Ex. 1004; Ex. 1011; Ex. 1012; Ex. 1013; Ex. 2044 ¶¶ 58–79, 134–140, 169–176).

Patent Owner also renews arguments (as argued over Ground 1) that the complexity (and cost) of combining a nucleic acid and FOBt test would have discouraged the ordinarily skilled artisan from combining the prior art as asserted by Petitioner. *Id.* at 44–45 (citing Ex. 2044 ¶¶ 131–139, 152–159, 161–176). Patent Owner argues that the ordinarily skilled artisan would have chosen to use just a single class of marker and just one type of testing. *Id.*

As for the addition of Kanaoka's teachings, Patent Owner's arguments are exactly the same for Petitioner's Ground 4 as they were for Ground 2. Resp. 47–51. We have already discussed these arguments above. *See supra* Section II.F.3.

3. Analysis

We have considered the parties' arguments and evidence and discuss critical aspects below. We agree with Petitioner's positions and adopt Petitioner's rationale as our own. We do not find Patent Owner's arguments sufficient to undermine Petitioner's more persuasive evidence and reasoning.

Petitioner asserts that the combination of Shuber and Vilkin teaches each limitation of claims 3, 4, and 14–19 and that it would have been obvious for a POSA to combine the fecal DNA assay of Shuber with the iFOBT assay of Vilkin to arrive at the claimed methods with a reasonable expectation of success. Pet. 41–48. Petitioner also asserts that Kanaoka further teaches the limitations of claims 12 and 13 and that a POSA would have had a reason to combine Kanaoka's RNA detection assay with Shuber and Vilkin for the same reasons discussed above. Pet. 48–49; *see supra* Section II.F. We agree.

When combined, Shuber and Vilkin and, where necessary, also Kanaoka, teach or suggest each step and element of claims 3, 4, and 12–19. As discussed above, Shuber and Vilkin together teach processing “freshly-collected” and not-frozen fecal matter by collecting it “at home,” by defecation into a sealable vessel, removing a portion of the sample and combining it with a buffer (in another sealable container) to preserve blood proteins, and combining the other portion of the sample with a stabilizing buffer (later claimed to preserve nucleic acids, e.g., DNA or RNA), to then

detect blood proteins and/or target epigenetically modified DNA (or RNA), to detect colorectal cancer (CRC). *See* Ex. 1001, 45:21–47:4. Kanaoka would have been combined with Shuber and Vilkin just as it would have been combined with Lenhard, Vilkin, and Itzkowitz, and the resultant teachings would be essentially the same. *See supra* Section II.F.

Unlike Lenhard, Shuber does not expressly combine FOBT and nucleic acid tests or suggest that combining the tests may result in increased sensitivity of detecting CRC. Petitioner relies on the statements discussed above in Lenhard—along with the similar teachings of Nishikawa, Kutzner, and Levin—to argue that the benefits of combining stool DNA and blood protein assays were well understood in and suggested by the prior art. Pet. 37–41 (citing, e.g., Ex. 1004, 143, 147, Table 4; Ex. 1011, 112; Ex. 1012, 34, 40; Ex. 1002 ¶¶ 227–239). Petitioner asserts that successfully combining the assays of Shuber and Vilkin “requires no more tha[n] the use of routine methods to perform [] a pair of well-established assays on separate portions of a fecal sample.” *Id.* at 41 (citing Ex. 1002 ¶ 238).

“[E]vidence of a motivation to combine need *not* be found in the prior art references themselves, but rather may be found in ‘the knowledge of one of ordinary skill in the art.’” *Dystar Textilfarben GmbH & Co. Deutschland KG v. C.H. Patrick Co.*, 464 F.3d 1356, 1366 (Fed. Cir. 2006). And, an obviousness “analysis ‘need not seek out precise teachings directed to the specific subject matter of the challenged claim, [to] take account of the inferences and creative steps that a person of ordinary skill in the art would employ.’” *Intel*, 61 F.4th at 1379–80 (quoting *KSR*, 550 U.S. at 418, 420). Moreover, the prior art combination does not need to be the best or preferred approach but need only be “desirable in light of the prior art.” *Honeywell*

Int'l Inc. v. 3G Licensing, S.A., 124 F.4th 1345, 1356 (Fed. Cir. 2025).

Finally, the “case law is clear that obviousness cannot be avoided simply by a showing of some degree of unpredictability in the art so long as there was a reasonable probability of success.” *Pfizer*, 480 F.3d at 1364.

We find on this record that Petitioner has shown that the combination of Shuber and Vilkin teaches or suggests each limitation of claims 3, 4, and 14–19 and that the combination of Shuber, Vilkin, and Kanaoka teaches or suggests each limitation of claims 12 and 13 for the reasons stated by Petitioner. Pet. 41–49; Ex. 1002 ¶¶ 240–277. As asserted by Petitioner, Shuber teaches a CRC screening test where a patient collects a fecal sample and seals it in a container with a nucleic acid stabilizing buffer solution, without requiring freezing, and that sample is shipped to a lab and tested via a hypermethylation assay or DNA mutation assay. *See* Ex. 1008, 1–4, 7, 10, 15. Shuber does not expressly say so, but its disclosed methods and materials suggest at-home collection by a patient. *See id.* at 3–4, 7, 10, 29 (discussing sample’s direct deposit into a sealable and shippable container, no freezing, stability for possibly weeks, decentralized collection and testing).

Vilkin also suggests at-home sample collection, and teaches that an off-the-shelf product/system can be used to provide an iFOBT CRC screen for patients where a small sample portion is separated from a larger fecal sample, sealed with an Hb stabilizer, and shipped without freezing to a lab for the testing. *See generally* Ex. 1005. That Vilkin teaches a method of specifically sampling over three days gives us no concern, as discussed above. *See supra* Section II.F.3.

Shuber provides high sensitivity for adenomas where Vilkin may not, Vilkin provides high sensitivity for “all significant colorectal neoplasia,” and it was well known that the two combined tests enabled detection of more cancer than either alone; thus, the two tests complement one another. Ex. 1005, 2523; Ex. 1008, 13–14; Ex. 1004; Ex. 1011–1013; Ex. 1002 ¶¶ 232–238; *see Outdry Techs. Corp. v. Geox S.p.A.*, 859 F.3d 1364, 1370–71 (Fed. Cir. 2017) (“Any motivation to combine references, whether articulated in the references themselves or supported by evidence of the knowledge of a skilled artisan, is sufficient to combine those references to arrive at the claimed process.”); *see also KSR*, 550 U.S. at 417 (“the predictable use of prior art elements according to their established functions” can reasonably be expected to succeed and is obvious).

Thus, having considered the parties’ respective arguments and evidence, we are persuaded that Petitioner has shown that the challenged claims would have been unpatentable over the asserted combinations of Shuber, Vilkin, and Kanaoka.

4. Summary and Conclusion

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 3, 4, and 14–19 are unpatentable as obvious over Shuber and Vilkin, and that claims 12 and 13 are unpatentable as obvious over Shuber, Vilkin, and Kanaoka. We do not find Patent Owner’s arguments sufficient to undermine Petitioner’s more persuasive evidence and reasoning.

H. OBVIOUSNESS OVER INBAR, VILKIN, AND SHUBER, AND ALSO ADDING KANAOKA (GROUNDS 6 AND 7)²²

We conclude above that all claims have been proven by a preponderance of the evidence to be unpatentable under Grounds 1–4, but for completeness, we analyze Petitioner’s Grounds 6 and 7 below.

Petitioner asserts that claims 3, 4, and 14–19, would have been obvious over Inbar, Vilkin, and Shuber (Ground 6). Pet. 56–65 (citing, *inter alia*, Ex. 1001, 18:33–37; Ex. 1002 ¶¶ 295–329, 232–236; Ex. 1005, *generally*, but also at 2519–22; Ex. 1008, *generally*, but also at 7:10–12, 7:19–21, 10:10–15, 14:1–5, 15:18–16:17; Ex. 1009, *generally*, but also at Abstr., 1:2–12, 3:48, 4:2–8, 4:16–5:22, 9:7–30, Figs. 1–4); *see also id.* at 49–55 (detailing Inbar’s disclosure relevant to independent claim 1—citing Ex. 1001, 18:33–37; Ex. 1002 ¶¶ 278–304; Ex. 1009, *generally*, but also at Abstr., 1:2–16, 3:3–8, 3:18–5:23, 6:4–32, 8:2–9:30, 10:1–15, Figs. 1–8). Petitioner also asserts that claims 12 and 13 would have been obvious over Inbar, Vilkin, Shuber, and Kanaoka (Ground 7). Pet. 65–66 (citing Ex. 1002 220–226, 220, 331; Ex. 1007 ¶¶ 21–26, 47–48, 59).

Patent Owner opposes Petitioner’s challenges. Resp. 45–64 (citing, *inter alia*, Ex. 1009, 2:31–3:3, 4:13–18; Ex. 2044 ¶¶ 112–128, 178). As it was under above-discussed Grounds 3 and 4, some of Patent Owner’s rationale in opposition to Petitioner’s Grounds 6 and 7 is generic to all

²² Under Ground 5, Petitioner asserts that Inbar anticipates claims 1 and 2. In view of Patent Owner’s disclaimer of these challenged claims, the Ground is moot. However, Petitioner’s rationale under Ground 5 relating to the disclosure of Inbar is relevant as it supports the unpatentability challenges under Grounds 6 and 7 for obviousness.

Grounds or are arguments made over Grounds 3 and 4; this has been addressed above and will not be revisited in detail.

Having considered the arguments and evidence presented by the parties, we determine that Petitioner has shown that the challenged claims would have been unpatentable as obvious over the cited references. We review the parties' positions and provide our analysis below.

1. Petitioner's Positions

Petitioner, under Grounds 6 and 7, asserts and cites to prior art references Shuber, Vilkin, and Kanaoka for essentially the same teachings and support as for Grounds 3 and 4. We will not rehash Petitioner's assertions in this regard.

Under Grounds 6 and 7, Petitioner adds to the Shuber-Vilkin and Shuber-Vilkin-Kanaoka prior art combinations the teachings of Inbar, which Petitioner asserts teaches a system specifically for collecting, portioning, and sealed storage of portioned fecal samples, each storage container having a respective preservation medium specific to the purpose of preserving the sample for specific testing. Pet. 56–57 (citing, *inter alia*, Ex. 1002 ¶¶ 306–309; Ex. 1009, Abstr., 1:2–12, 3:48, 4:2–8, 4:18–5:22; 9:7–30, Figs. 1–4). Petitioner asserts that when the sample portioning, preserving, storing, and shipping for iFOBT and nucleic acid assay CRC screening of the Vilkin-Shuber combination is considered (as under Grounds 3 and 4), it would have been advantageous and obvious to a person of ordinary skill in the art to use a system and apparatus as taught by Inbar to collect, preserve, store, and ship such samples because Inbar's system is specifically made for such a process. *See id.*; *see also* Ex. 1002 ¶¶ 278–304 (analyzing how Inbar discloses a system for practicing claim 1's method, a prelude to shipping the samples

for lab testing per claim 3's method). Thus, Petitioner asserts that there was a reason to combine the prior art and there would have been a reasonable expectation of success in doing so. *See* Pet. 57.

Petitioner maps the teachings of Inbar, Shuber, and Vilkin to the steps and elements of claims 3, 4, and 14–19, and adds the teachings of Kanaoka for the elements of claims 12 and 13. Again, Petitioner asserts Shuber, Vilkin, and Kanaoka in essentially the same ways as for Grounds 3 and 4 (*see supra* Section II.F–G); thus, we summarize the ways Petitioner asserts Inbar as an addition to the already-addressed prior art combinations.

Regarding claim 3, Petitioner asserts that Inbar teaches all steps and elements recited by independent claim 1, which is incorporated into claim 3 by its dependency, by disclosing “a system for collecting multiple specimens from a stool sample into separate sealable containers, each containing its own medium for preserving a chemical entity or living matter, so that the multiple specimens can be shipped to a laboratory where different assays are performed on the separately preserved specimens.” Pet. 49–55, 58–63 (citing, *inter alia*, Ex. 1002 ¶¶ 278–304, 311–317, 320–; Ex. 1009, Abstr., 1, 3–6, 9–10, Figs. 1–4, 6–8). Petitioner asserts that Inbar teaches a toilet-seat collection device (suggesting and teaching at-home collection, particularly in view of Shuber and Vilkin), respective sealable containers and sampling devices for storing portions of the fecal sample (for shipping to a test lab), respective preservation medium in each container (to preserve DNA per Shuber and preserve blood protein per Vilkin), and testing (for, *inter alia*, blood and DNA markers in the specimen, in context of Shuber teaching DNA testing and Vilkin teaching iFOBT), all without freezing the sample (particularly in view of Shuber and Vilkin). *Id.*

Overall, in summary, Inbar is asserted to teach a device or system that particularly suits the combined sample collection, portioning, and testing taught by Shuber, Vilkin, and Kanaoka. *See supra* Sections II.F–G. Inbar is not asserted to necessarily teach any other elements of claims 4 and 12–19 not taught by Shuber, Vilkin, and Kanaoka. *See* Pet. 63–66.

Having summarized Petitioner’s positions on the prior art combinations and their teaching of the claimed subject matter, we turn to Patent Owner’s positions in opposition.

2. Patent Owner’s Positions

Patent Owner renews prior arguments: Shuber and Vilkin would not have been combined; two different preservation solutions were not required for nucleic acids and proteins, which means Inbar’s two separate containers would not be needed; Kanaoka requires freezing and there was no reason to add its teachings to improve the other reference’s tests. Resp. 45–51 (citing, *inter alia*, Ex. 1009, 4; Ex. 2044 ¶¶ 116, 123–124). Patent Owner also argues that Inbar does not mention at-home sample collection and, rather than the DNA and iFOBT tests of the other prior art, Inbar teaches testing for bacteria and parasites. *Id.* at 46 (citing Ex. 1009, 2–4; Ex. 2044 ¶¶ 112–128, 178).

3. Analysis

We have considered the parties’ arguments and evidence and, again, we agree with Petitioner’s positions and adopt Petitioner’s rationale as our own. We do not find Patent Owner’s arguments sufficient to undermine Petitioner’s more persuasive evidence and reasoning. Because we discussed above why Shuber, Vilkin, and Kanaoka would have been combined by the person of ordinary skill in the art with a reasonable expectation of success

and teach each step or element of the challenged claims, we will not rehash our positions. We focus on Petitioner's assertions regarding Inbar and Patent Owner's arguments specific to these Grounds.

Petitioner asserts that the combination of Inbar, Shuber, and Vilkin teaches each limitation of claims 3, 4, and 14–19 and that it would have been obvious for an ordinarily skilled artisan to have combined the fecal specimen testing preparation system of Inbar with the DNA and blood protein assays of Shuber and Vilkin, respectively, to arrive at the claimed methods with a reasonable expectation of success, and also that Kanaoka further teaches the limitations of claims 12 and 13 and that the skilled artisan would also have had a reason to combine Kanaoka's RNA detection assay with Inbar, Shuber and Vilkin for the same reasons discussed above. We agree. Adding Inbar's system and components to the other cited prior art would simply make the combination more streamlined and efficient because Inbar provides the two containers, sampling/portioning tools, and respective preservatives required once Shuber's DNA testing and Vilkin's iFOBT (and Kanaoka's RNA testing) are used together.

In support of this conclusion, we adopt as our own Petitioner's rationale and cited evidence. Petitioner has shown that the combination of Inbar, Shuber, and Vilkin teaches or suggests each limitation of claims 3, 4, and 14–19 and that the combination of Inbar, Shuber, Vilkin, and Kanaoka teaches or suggests each limitation of claims 12 and 13 for the reasons stated by Petitioner. Pet. 56–66; Ex. 1002 ¶¶ 311–331.

Patent Owners arguments directed to Grounds 6 and 7, as well as Patent Owner's over-arching arguments discussed above, are not persuasive.

Regarding Patent Owner's new arguments, although Inbar does not expressly mention at-home sample collection, we find Inbar suggests it. Inbar's flushable toilet collection system, sampling device(s), sealable containers, and preservative mediums are designed for ease of use by a patient, and for containment of a sample for shipping to a lab for analysis, without the need for freezing. Ex. 1009, 1–7, Figs. 1–8. Even without mentioning the patient's home, these disclosures suggest that all steps before lab analysis occur there, particularly in view of Vilkin's and Shuber's disclosures suggesting the same (discussed above).

Although Inbar is, indeed, most focused on testing fecal samples for bacteria and parasites and preserving such analytes for testing, Inbar does mention that fecal testing can include “detection of cryptic blood” and “preparation of a biochemical profile.” Ex. 1009, 1. Thus, Inbar suggests that its system and devices can be used for the DNA and iFOBT testing taught by Shuber and Vilkin.

Thus, having considered the parties' respective arguments and evidence, we are persuaded that Petitioner has shown that the challenged claims would have been unpatentable over the asserted combinations of Inbar, Shuber, Vilkin, and Kanaoka.

4. Summary and Conclusion

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 3, 4, and 14–19 are unpatentable as obvious over Inbar, Shuber, and Vilkin, and that claims 12 and 13 are unpatentable as obvious over Inbar, Shuber, Vilkin, and Kanaoka. Patent Owner's arguments to the contrary are insufficient to outweigh the evidence and reasoning Petitioner provides.

III. CONCLUSION²³

On the record before us, Petitioner demonstrates by a preponderance of the evidence that claims 3, 4, and 12–19 of the '746 patent would have been obvious over the asserted prior art, under each asserted ground. Accordingly, all challenged claims of the '746 patent are unpatentable under 35 U.S.C. § 103.

Our final decision is summarized as follows:

Claim(s)	35 U.S.C. §	Reference(s)/ Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
3, 4, 14–19	103	Lenhard, Vilkin, Itzkowitz	3, 4, 14–19	
12, 13	103	Lenhard, Vilkin, Itzkowitz, Kanaoka	12, 13	
3, 4, 14–19	103	Shuber, Vilkin	3, 4, 14–19	
12, 13	103	Shuber, Vilkin, Kanaoka	12, 13	
3, 4, 14–19	103	Inbar, Vilkin, Shuber	3, 4, 14–19	
12, 13	103	Inbar, Vilkin, Shuber, Kanaoka	12, 13	
Overall Outcome			3, 4, 12–19	

²³ If Patent Owner wishes to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to this Decision, see the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). Patent Owner has a continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

ORDER

Accordingly, it is hereby:

ORDERED that, pursuant to 35 U.S.C. § 314, per *inter partes* review of claims 3, 4, and 12–19 of U.S. Patent 11,970,746 B2, the challenged claims are unpatentable; and

FURTHER ORDERED that, as this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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